

The role of nanotechnology and artificial intelligence in optimizing thermal energy systems

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HIGHLIGHTS

- Integrates nanotechnology and AI to optimize thermal energy system performance.
- Analyses nano-enhanced PCMs and nanofluids for energy storage and heat transfer.
- Reviews AI algorithms for predictive control and efficiency in thermal systems.
- Identifies research gaps in scalability, cost, and material sustainability.
- Proposes future research roadmap for AI-nano hybrid energy solutions.

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ABSTRACT

The growing demand for clean energy and the limitations of conventional thermal systems necessitates the integration of advanced technologies to enhance efficiency, adaptability, and sustainability. This review critically examines recent advancements in the application of nanotechnology and artificial intelligence for optimizing thermal energy systems, including solar collectors, heat exchangers, and latent heat storage units. Nanotechnology (particularly the use of nano-enhanced phase change materials and nanofluids such as Al_2O_3 and CuO) has shown to improve thermal conductivity by up to 28.8 %, accelerating energy absorption and storage rates. Concurrently, artificial intelligence algorithms, especially artificial neural networks and particle swarm optimization, enable predictive modelling, real-time system control, and fault detection, with some models achieving prediction accuracies above 97 % under complex operational conditions. The review emphasizes the synergistic potential of combining these technologies to create intelligent, self-regulating thermal energy systems. However, the paper also identifies critical challenges including computational overhead, cost of nanoparticle synthesis, lack of reproducibility in artificial intelligence implementations, and insufficient validation under extreme scenarios. Commercial deployment case studies (such as artificial intelligence-driven phase change material-based heating, ventilation, and air conditioning systems in smart buildings) are discussed to illustrate practical viability, reporting energy savings of up to 28 % with return-on-investment periods under three years. The

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paper concludes by proposing integrated research directions that combine multiscale material innovation with robust artificial intelligence training on dynamic datasets. This dual approach is essential to developing scalable, cost-effective, and resilient thermal energy systems capable of supporting global energy transitions.

Nomenclature			
<i>Abbreviation</i>		ML	Machine Learning
ACO	Ant Colony Optimization	MLP	Multi-layer Perceptron
AI	Artificial Intelligence	MPC	Model Predictive Control
ANFIS	Adaptive Neuro-Fuzzy Inference System	MSE	Mean Squared Errors
ANN	Artificial Neural Network	MWCNT	Multi-Walled Carbon Nanotube
BPNN	BackPropagation Neural Network	M-GPR	Gaussian Processing Regression Model
CPSS	Cyber-Physical-Social Systems	NePCM	Nano-enhanced Phase Change Material
CSA	Cuckoo Search Algorithm	ODNN	Optimized Deep Neural Network
DL	Deep Learning	PCM	Phase Change Material
DRL	Deep Reinforcement Learning	PSO	Particle Swarm Optimisation
DQNs	Deep Q-Networks	PV	Photovoltaic Panels
EEPS	Energy and Electric Power Systems	PV/T	Photovoltaic/Thermal
FFNN	Feedforward Neural Networks	PI	Prediction Interval
FM	Forecasting Model	QRF	Quantile Random Forest
FPSC	Flat Plate Solar Collector	RAE	Relative Absolute Error
GA	Genetic Algorithms	RBF	Radial Basis Function
GAN	Generative Adversarial Network	RF	Random Forest
GBR	Gradient Boosting Regressor	RL	Reinforcement Learning
GM	Generalised Modelling	RMSE	Root Mean Square Error
GOA	Grasshopper Optimisation Algorithm	SA	Simulated Annealing
GPR	Gaussian Process Regression	SL	Supervised Learning
HVAC	Heating, Ventilation, and Air Conditioning	SMR	Small Modular Reactor
IoT	Internet of Things	SSL	Semi-Supervised Learning
ITES	Ice Thermal Energy Storage	SVR	Support Vector Regression
LHTES	Latent Heat Thermal Energy Storage	SWHS	Solar Water Heating System
LSTM	Long Short-Term Memory	SWCNT	Single-Walled Carbon Nanotube
KNN	K-Nearest Neighbours	TES	Thermal Energy System
KPI	Key Performance Indicator	TIC	Theil Inequality Coefficient
MAE	Mean Absolute Error	VAE	Variational Autoencoder
MAPE	Mean Absolute Percentage Error	WNN	Wavelet Neural Networks
		XGBoost	Extreme Gradient Boosting

1. Introduction

With escalating environmental concerns, the rising worldwide energy demand has created an urgent need for more efficient and sustainable energy systems [1]. Fossil fuels dominate the energy sector and are rapidly diminishing, and have considerable detrimental effects on the environment, exacerbating climate change via greenhouse gas emissions [2]. This has compelled civilizations to pursue alternate energy sources that are both renewable and more ecologically sustainable. Renewable energy sources such as solar, wind, and hydropower have become prominent because of their capacity to reduce environmental pollution substantially. Nonetheless, the intrinsic problems of these energy sources, like intermittency and reduced energy densities, need the development of sophisticated technology to enhance their efficiency and viability. Within this framework, two essential technologies have arisen: nanotechnology and Artificial Intelligence (AI) [3]. These technologies provide potential solutions to improve the efficiency and reliability of Thermal Energy Systems (TESs). This analysis examines the contributions of nanotechnology and AI in enhancing TESs, emphasizing energy efficiency and sustainability improvements [4].

TESs are important in the utilization of renewable energy resources. They convert energy from renewable sources like solar or wind into thermal energy for a number of uses, like electricity generation, heating,

and cooling [5]. Solar energy is the most common and widely utilised kind of renewable energy. Solar thermal systems, such as the Solar Water Heating System (SWHS) and the Flat Plate Solar Collector (FPSC), on the other hand, are systems that are used to harvest and store solar energy for use later [6]. These systems are important in the reduction of fossil fuel dependency as they provide a clean source of energy. Another challenge that TESs face is how to achieve maximum efficiency. As a renewable energy source, solar power is inconsistent and has low energy density compared to fossil fuels, thus the a need for energy storage and conversion for successful implementation [7]. FPSCs are extensively used for water heating applications. These devices trap solar radiation and transmit the absorbed thermal energy to a working fluid, such as water or air [8].

Notwithstanding its prevalent use, the thermal efficiency of FPSCs is often constrained by elements such as heat losses and the intrinsic characteristics of the working fluids. Researchers have investigated many approaches to improve solar collectors' thermal efficiency, including using nanofluids as working fluids [9]. The heat transfer efficacy of solar collectors can be significantly improved by nanofluids, which are fluids that are infused with nanoparticles that exhibit exceptional thermal conductivity [10]. Research has demonstrated that the integration of nanofluids, such as Al_2O_3 , CuO , and TiO_2 , into FPSCs can substantially enhance the efficacy of the overall system and the rate of heat transfer. Tiwari et al. [11] demonstrated that the thermal

efficiency of FPSCs was improved by 31.64 % when Al_2O_3 nanofluid was applied at a volume concentration of 1.5 %. The thermal efficiency of FPSCs was increased by 5 % by a hybrid nanofluid that contained silver, graphite, and carbon nanotubes, as discovered by Otanicar and Golden. These findings emphasize nanotechnology's ability to supplant the limitations of traditional TESs and improve their overall efficiency [12].

Energy efficiency is one of the critical aspects that define the design and operation of TESs. Organized energy systems reduce the amount of energy used and minimize environmental impacts through the reduction of greenhouse gas emissions [13]. Improving the effectiveness of TESs is an important task to increase the energy output and minimize energy losses. Optimization is critical in renewable energy because the sources are inherently stochastic and random, such as solar and wind [14]. Numerous optimization strategies have been devised to improve overall system performance, including both passive and active approaches. Passive techniques, including optimizing design and materials in energy systems, do not need external energy input [15]. Conversely, active solutions use external forces, such as pumps or fans, to enhance heat transmission. One of the most promising optimization strategies is nanotechnology, which has shown the ability to raise the thermal conductivity of working fluids and improve heat transfer efficiency [16].

Nanotechnology has transformed TESs by facilitating the creation of materials with enhanced thermal characteristics. Nanofluids have been thoroughly investigated for their capacity to enhance heat transfer rates in thermal systems [17]. The increased thermal conductivity of nanoparticles facilitates enhanced heat absorption and transmission, resulting in improved thermal efficiency. Moreover, the application of nanofluids has demonstrated improved stability in thermal systems by reducing sedimentation and enhancing the long-term performance of heat exchangers [18]. In addition to advancements in nanotechnology, AI is emerging as a powerful tool for optimizing the performance of TESs [19].

Artificial Neural Networks (ANNs) and Particle Swarm Optimisation (PSO) have been applied for the emulation and prediction of the performance of energy systems under various operational conditions. Such approaches help build more accurate and robust energy systems with the help of processed data and predictive algorithms [20]. The application of materials on the nanoscale to improve the attributes of such materials is what is known as nanotechnology. Applications of TESs that make use of nanofluids and other nanomaterials have been extensively documented in the scientific literature as a result of their utilization of nanotechnology. Increasing the thermal conductivity of working fluids and the heat transfer coefficient has been accomplished by the use of nanofluids, which are a mixture of nanoparticles and base fluids [21]. This makes them ideal candidates for improving the performance of the solar thermal systems, and other heat exchange systems. A detailed analysis of the applicability of nanofluids in solar water heaters and FPSCs has been made. Scientists have demonstrated that by adding nanoparticles, like Al_2O_3 , CuO, and TiO_2 , into the standard working fluids, heat transfer coefficients and system performance may improve. Yousefi et al. [22] found that the thermal efficiency was increased by 28 % with Al_2O_3 nanofluid in FPSCs, and 35 % using MWCNT nanofluids. As seen from these results, nanotechnology holds great promise for improving and optimally utilizing TESs. On the other hand, AI has been increasingly utilised in the process of optimizing TESs [23].

Modelling, simulating, and forecasting the performance of energy systems have all been accomplished through the utilization of AI approaches. These methodologies include ANNs, PSO, and Adaptive Neuro-Fuzzy Inference Systems (ANFIS). Through the examination of huge datasets and the recognition of patterns that would be difficult to discover using conventional methods, these approaches offer a strategy that is both more exact and effective in terms of improving system performance. Kalani et al. [24] presented a model for evaluating the efficiency of a PV/T solar collector utilizing ZnO/water nanofluid through ANNs and particle swarm optimization. Applying 130 experimental data sets, they showed that both ANFIS and Radial Basis

Function (RBF) methods provide the highest performance for predicting the outlet temperature as well as the total system efficiency. Al-Waeli et al. [25] similarly evaluated the performance of PV/T systems using NePCM and nanofluid, indicating that AI-based models yielded dependable estimations of system efficiency. Fig. 1 shows an overview of the workflow of thermal energy storage systems in combination with AI techniques.

Unlike earlier models of AI, wherein an apparent assistive role was done to energy systems, AI has come of age since it has graduated to becoming the main driver of smart control, forecasting, and strategy optimization. Hot applications have emerged in building and industrial thermal regulation, where Heating, Ventilation, and Air Conditioning (HVAC) systems represent a considerable fraction of energy consumption. The application of real-time thermal load forecasting, fault detection, and dynamic energy management has made much use of AI algorithms, especially ML, ANNs, and ANFIS [26].

One such application is where predictive maintenance strategies have been devised with AI in-built techniques so that faults are detected before the eventual failure of systems. This strategy will minimize the associated costs of maintenance and the idle times of operation. Intelligent control systems automatically recalibrate the HVAC functions by accessibility modes, weather forecasts, and indoor conditions, for example, setting thermal comfort at energy-minimizing levels. These forms now establish a new profile of TES' effectiveness in smart buildings.

A recent article published by Ukoba et al. (2024) [26] shows how, in hybrid AI models where Deep Learning (DL) techniques are coupled with other fault classification algorithms, these models could significantly enhance system reliability and decrease energy usage by proactive diagnostics. Their work is an example of increasingly shaping the synergy of AI techniques with actual physical system modelling to connect data-driven intelligence with pragmatic energy engineering.

The success of AI in building environments creates a possibility of greater generalizability across other TESs, such as solar collectors, energy storage devices, heat exchangers, and so on. Induction of AI's capabilities in pattern-recognition, performance-prediction, and autonomy from control could leverage these systems even further when combined with the modern advanced materials or nanofluids and Nano-enhanced Phase Change Materials (NePCMs). Real-world implementations have substantiated this potential. For instance, AI-powered HVAC systems have achieved energy savings of up to 28 % through predictive control and fault detection mechanisms, with return-on-investment periods under three years [26]. Similarly, in photovoltaic thermal (PV/T) systems, hybrid AI models integrating deep learning with fault diagnostics have demonstrated improved reliability and reduced operational energy consumption [27]. These concrete outcomes highlight how AI is not merely theoretical but is actively transforming thermal energy systems through measurable gains in efficiency, cost savings, and operational resilience. Thus, AI integration in thermal sciences establishes a practical and data-driven path toward sustainable energy innovation.

1.1. Background and motivation

The escalating global energy demand, coupled with pressing environmental concerns, has intensified the need for sustainable and efficient TESs. Traditional fossil fuel-based systems are unsustainable, prompting a shift toward renewable energy integration. However, renewable sources like solar and wind suffer from intermittency, necessitating advanced solutions for energy storage and heat transfer optimization.

1.2. Current challenges and research gaps

Despite advancements, existing TESs face critical limitations: (1) low thermal conductivity of conventional PCMs, (2) inefficiencies in heat

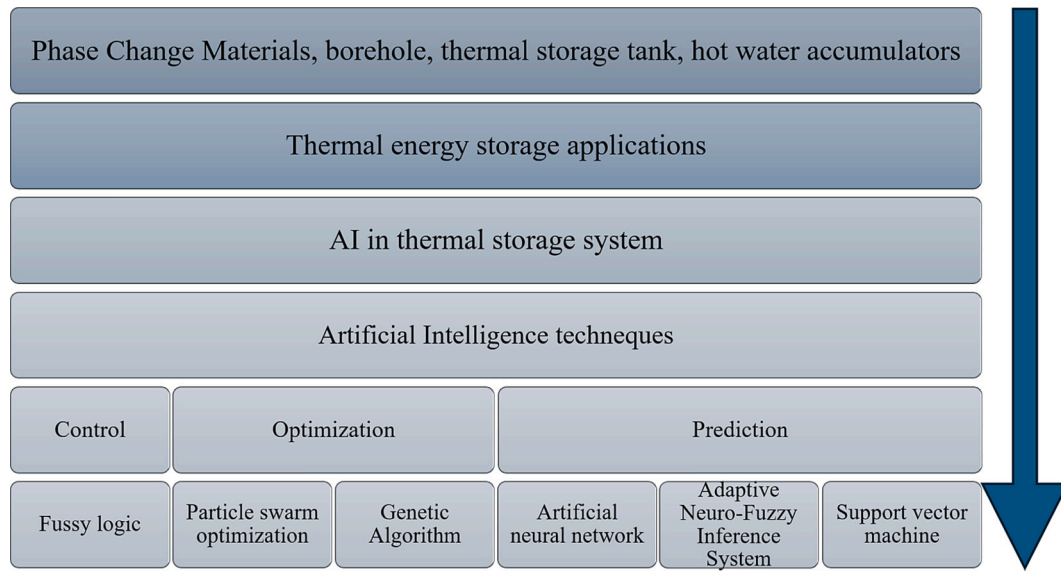


Fig. 1. Combination of the AI techniques with the thermal energy storage systems.

exchangers and solar collectors, and (3) a lack of real-time adaptive control. While nanotechnology (e.g., nanofluids, nano-enhanced PCMs) improves thermal properties, scalability and cost remain hurdles. Similarly, AI-driven predictive models show promise but are often siloed from material innovations.

1.3. Novelty and contribution of this work

This study bridges these gaps by synergizing nanotechnology and AI to create intelligent, self-optimizing thermal systems. Key innovations include:

- **Material Advancements:** Nano-enhanced PCMs and hybrid nanofluids to boost heat transfer and storage capacity.
- **AI Integration:** Machine Learning (ML) for real-time performance prediction, adaptive control, and fault detection.
- **System-Level Optimization:** A unified framework combining nanomaterial properties with AI-driven decision-making for dynamic energy management.

By integrating these domains, we demonstrate a transformative approach to enhance efficiency, reduce costs, and enable scalability, addressing a critical gap in the literature.

1.4. Methodology

This review adopts a narrative approach to synthesize the recent advancements in the application of nanotechnology and AI for optimizing TESs. Although not a systematic review, we employed a structured process to ensure comprehensiveness and relevance.

Literature Sources:

Peer-reviewed articles were retrieved from major scientific databases including Scopus, Web of Science, ScienceDirect, IEEE Xplore, and SpringerLink. Only English-language publications were considered. The search strategy used Boolean combinations of terms such as: NePCM, thermal conductivity, latent heat, nanofluid, hybrid nanofluid, solar collector, heat exchanger, AI, ML, DL, TESs, HVAC, and energy optimization.

Inclusion Criteria:

- Studies published between 2013 and 2024.

- Focused on thermal energy storage, solar thermal systems, nanomaterials in PCMs, or AI-based modelling/control in energy systems.
- Empirical studies with quantitative results or review papers providing comparative analysis.

Exclusion Criteria:

- Articles lacking technical detail or experimental validation.
- Non-peer-reviewed reports, patents, or conference abstracts with insufficient data.

This process resulted in the selection of approximately 180 core publications, which were further analysed and categorized based on thematic relevance and technical contribution.

2. Fundamentals of thermal energy systems

The principles of TESs concentrate on techniques for storing and transporting energy as heat. These systems are classified into active and passive configurations. Active systems often use mechanical elements such as pumps and fans to control heat transmission, while passive systems depend on natural convection or conduction without additional energy sources. Both methodologies seek to minimize energy use, improve efficiency, and mitigate environmental effects. The use of nanotechnology and sophisticated materials has become crucial in improving the performance of TESs [28].

The related methodology incorporates a detailed exploration of approaches for coupling nanoscale phenomena, such as Brownian motion and quantum effects, with system-level AI architectures. This integration is paramount for developing predictive models that accurately reflect the complex interplay between microscopic behaviours and macroscopic system performance in thermal energy applications. The techniques for multiscale modelling have been delineate, where molecular dynamics simulations or direct numerical simulations of nanoscale fluid dynamics, incorporating phenomena like Brownian motion and inter-nanoparticle forces, generate data. This data then serves as input for training AI models, including neural networks and Reinforcement Learning (RL) algorithms, which operate at the system level. These AI architectures learn to extrapolate nanoscale behaviours to predict overall system efficiency, heat transfer rates, and long-term stability. Furthermore, the development of hybrid AI models that directly embed physical laws governing nanoscale phenomena, thereby enhancing the

interpretability and robustness of system-level predictions has been discussed. This approach allows for a more nuanced understanding of how alterations at the atomic or molecular scale translate into measurable improvements or challenges in large-scale TESs.

2.1. Fundamentals of thermal energy systems

2.1.1. Active thermal energy systems

Active TESs use mechanical or electrical components to improve heat storage, transport, and usage. In contrast to passive systems that depend on natural heat transfer, active systems use pumps, fans, compressors, and other gear to control the movement of thermal energy more effectively [29]. A vital element of active TESs is the heat exchanger, which enables the transfer of heat between two or more fluids at varying temperatures. Heat exchangers are often used in HVAC systems and solar water heaters to absorb, store, and transfer thermal energy [29]. In solar thermal systems, heat exchangers convey the heat captured by solar collectors to a fluid, usually water or a glycol mixture, which may subsequently be stored or used for heating buildings or water. The efficacy of this process depends on the heat exchanger's design, the fluid's characteristics, and the operating parameters. Nanofluids, fluids containing suspended nanoparticles, are used in several systems to augment thermal conductivity and heat transfer rates, increasing overall performance [30].

A crucial component of active TESs is thermal storage technology, which accumulates surplus thermal energy during periods of demand surpassing supply. There are three primary mechanisms employed in these systems: sensible heat storage, latent heat storage, and thermochemical energy storage [31]. Latent heat storage, often using PCMs, has become prominent owing to its elevated energy storage density and isothermal properties. In active systems, thermal energy storage may stabilize the energy supply in renewable systems characterized by fluctuating energy input. In concentrated solar power plants, surplus solar energy may be retained during peak sunshine hours and then released during times of less sunlight, providing a continuous energy supply [32]. The regulatory mechanisms of active TESs are essential for enhancing their efficiency. Advanced control algorithms, often integrating AI and ML methodologies, may oversee, forecast, and modify system parameters in real-time to attain optimal efficiency.

AI-driven systems may forecast future energy demand and appropriately modify heating or cooling rates, thus minimizing energy waste. These intelligent controls are especially advantageous in systems influenced by external variables, such as meteorological conditions or building occupancy, that impact thermal demand. Predictive maintenance is a burgeoning use of AI in active systems, allowing operators to foresee faults and enhance maintenance plans, augmenting system dependability and durability [33]. A difficulty with active TESs is the energy required to operate mechanical components, which can offset the energy savings achieved through enhanced thermal management. Advancements in energy-efficient components and the incorporation of renewable energy are contributing to a reduction in the operating energy costs of these systems [34]. Solar Photovoltaic Panels (PV) may energize the mechanical components of solar thermal systems, enhancing the overall sustainability of the configuration. Furthermore, advancements in materials, including nano-enhanced PCMs, are consistently enhancing the energy storage capacity and thermal conductivity of these systems, thereby increasing their efficiency and cost-effectiveness [35].

2.1.2. Passive thermal energy systems

Passive TESs are designed to regulate heat fluxes in buildings and other structures without mechanical components, using the inherent principles of heat transmission. They use the intrinsic characteristics of materials and ambient variables to achieve effective thermal control. In structures, passive thermal techniques diminish heating and cooling requirements by systematically absorbing and releasing heat. These

systems optimize energy use and decrease dependence on active thermal systems, including HVAC systems, enhancing environmental sustainability [36]. Thermal mass is a critical element of passive TESs, as it refers to a material's capacity to absorb, store, and discharge heat over time. Significant thermal mass is possessed by materials such as concrete, brick, or stone, which allows them to absorb heat during the day and progressively release it at night. This helps to regulate indoor temperatures by reducing volatility and improving comfort, while simultaneously reducing the need for active heating or cooling systems.

Thermal mass is often integrated with deliberate architectural design, including the orientation of buildings to optimize or reduce sun exposure according to the season [37]. Insulation is a crucial element of passive thermal systems. Insulation regulates heat transfer between a building's interior and outside, stabilizing inside temperatures. Well-insulated structures may save winter heat and maintain lower summer temperatures, reducing the need for active thermal regulation systems [38]. Insulation in walls, roofs, and floors often employs materials with low heat conductivity, like foam, fiberglass, or cellulose. Effective insulation boosts energy efficiency and improves occupant comfort by establishing a more stable temperature environment [39]. Besides thermal mass and insulation, passive systems use solar gain and natural ventilation to control internal temperatures. Solar gain denotes the absorption of sunlight to heat a structure. In colder areas, structures are often constructed with expansive south-facing windows that let sunshine penetrate during winter, heating the inside. Shading devices, such as overhangs or blinds, help mitigate warming in summer by obstructing excessive sunlight.

Passive thermal systems may optimize solar gain by using natural energy sources for heating, hence reducing reliance on artificial systems [40]. Natural ventilation is a vital component of passive TESs, using airflows to control temperature and air quality without mechanical intervention. Well-engineered ventilation systems facilitate the ingress of cold air into a structure while evacuating heated air, reducing the need for air conditioning [41]. Cross-ventilation, which utilizes apertures on opposing sides of a structure to promote circulation, and stack ventilation, which leverages variations in air pressure to facilitate fresh air, are often used to improve thermal comfort. These strategies are particularly effective in temperate settings characterized by moderate outdoor temperatures [42]. Integrating thermal mass, insulation, solar gain, and ventilation may enhance a building's energy efficiency via passive TESs. The effectiveness of these systems relies on meticulous design and consideration of local environmental factors. In some instances, further techniques may be required to enhance energy efficiency, such as using high-performance glass or integrating shading components. Although straightforward, passive systems sometimes need meticulous design and exact engineering to guarantee their successful operation across all environmental conditions [37].

2.2. Key performance indicators for optimization

Key Performance Indicators (KPIs) are crucial for assessing and enhancing TESs. These metrics provide a systematic method to evaluate system performance, enabling efficiency, sustainability, and cost-effectiveness enhancements. In TES, KPIs are often classified into thermal efficiency, energy storage capacity, reaction time, economic feasibility, and environmental effect, enabling engineers and designers to pinpoint areas for improvement and evaluate various solutions. By comprehending and enhancing these KPIs, thermal systems may be refined for diverse applications, including building heating, cooling, and industrial activities [43]. Table 1 presents the selected KPIs for nanotechnology and AI in enhancing TES.

Thermal efficiency is a critical KPI, reflecting the system's capacity to transform and use heat energy efficiently. High thermal efficiency indicates that greater input energy is transformed into beneficial heat or cooling, minimizing waste and enhancing the system's total energy performance [44]. Thermal efficiency may be quantified using

Table 1
KPIs for Nanotechnology and AI in Enhancing TESs.

Category	KPI	Description
Efficiency	Thermal efficiency	The ratio of useful thermal energy output to total thermal energy input.
	Energetic efficiency	The ratio of useful exergy output to total exergy input.
	Heat transfer enhancement	Improvement in heat transfer rate compared to conventional methods.
	Temperature uniformity	Consistency of temperature distribution within the system.
	Heat loss reduction	Minimization of heat loss to the environment.
Cost-Effectiveness	Initial investment	Cost of implementing nanotechnology and AI solutions.
	Operational costs	Energy consumption, maintenance, and repair expenses.
	Return on investment (ROI)	Economic benefits are realized compared to the initial investment.
	Payback period	Time is required to recoup the initial investment.
	Life cycle cost analysis	Total cost of ownership over the system's lifetime.
Sustainability	Environmental impact	Reduction in greenhouse gas emissions, water consumption, and waste generation.
	Resource utilization	Efficient use of materials and energy.
	Reliability	System uptime and operational stability.
	Durability	Longevity and lifespan of the system.
	Social impact	Benefits to communities and society include job creation and improved quality of life.

characteristics like heat transfer rates, temperature differentials, and environmental heat loss. Enhancing thermal efficiency necessitates optimizing heat exchanger designs, selecting suitable materials with elevated thermal conductivity, and implementing techniques to reduce thermal losses during operation [45]. Energy storage capacity is a KPI, particularly for thermal energy storage systems. This statistic assesses the system's capacity to retain thermal energy for subsequent use, which is crucial for scenarios with variable energy demand. Systems that include solar thermal energy depend on effective energy storage to provide heat during times devoid of sunshine [46].

The energy storage capacity is determined by the characteristics of the materials used in the system, including their heat capacity, phase change temperature, and energy density. Improving energy storage capacity may provide superior load control and increased resilience against supply disruptions [47]. The response time, or the system's capacity to adapt to fluctuations in energy demand, is another critical KPI. In several applications, especially in dynamic situations such as buildings or industrial settings, the TES must promptly react to variations in temperature or energy demands. A reduced reaction time signifies that the system may swiftly modify its heat output, enhancing overall efficiency and user comfort [48]. Response time is influenced by the system's thermal inertia, the velocity of heat transfer mechanisms, and the management techniques used to manage energy flow. Advanced control algorithms and intelligent sensors may enhance reaction times, guaranteeing that the TES adjusts rapidly to real-time fluctuations [49].

Economic viability is another KPI, integrating early investment expenditures with long-term operations costs. This statistic assesses the financial viability of a TES by juxtaposing installation expenses, maintenance obligations, and energy savings during the system's lifespan. For TES to achieve widespread adoption, particularly in commercial and residential structures, it must provide a cost-effective alternative to conventional heating and cooling systems [50]. Economic feasibility may be enhanced using resilient, low-maintenance materials, engineering systems that need little energy consumption, and incorporating renewable energy sources to mitigate operating expenses. Furthermore, lifetime cost analysis may provide an in-depth comprehension of various TES technologies' financial advantages and compromises [51]. The

environmental effect is becoming recognized as a critical KPI due to the intensifying worldwide emphasis on sustainability. TES is often integrated into comprehensive programs to diminish carbon emissions, enhance energy efficiency, and shift toward renewable energy sources [52].

The environmental impact KPI evaluates aspects like greenhouse gas emissions, resource use, and the risk of pollution or ecological deterioration. Systems using renewable energy sources or materials with little environmental impact often get superior ratings in this area. Enhancing TES for ecological effect entails using sustainable materials, reducing waste heat, and engineering systems compatible with renewable energy sources like solar or wind power [53]. Additional KPIs for material performance have been made available as a result of the incorporation of nanotechnology into TES. These KPIs include the thermal conductivity of nanofluids and the latent heat storage capacity of nano-enhanced PCMs. In order to increase the thermal transfer characteristics and energy storage efficiency of thermal energy storage, these cutting-edge materials have been developed. If successful, this might lead to significant enhancements in performance [46]. PCMs that have been upgraded with nanotechnology have the potential to store more energy per unit volume than traditional materials. This would result in an increase in the energy density and compactness of thermal energy storage. The use of nanoparticles, on the other hand, raises new concerns, such as the requirement to guarantee the durability and stability of these materials over extended periods of operation. These concerns need to be overcome in order to totally improve the performance of the system [54].

In AI and ML, KPIs of system intelligence and adaptability are also developing. Intelligent thermal energy storage systems using AI algorithms can forecast energy demand, optimize energy storage and distribution, and modify operating parameters in real-time [33]. These systems are assessed on their capacity to learn from previous data, predict future energy requirements, and reduce energy waste via accurate control. KPIs in this arena include predictive accuracy, energy savings realized via AI optimization, and system flexibility in fluctuating situations. As AI progresses, these KPIs will gain significance for enhancing the next generation of TES [55].

2.3. Challenges in current thermal energy systems

Contemporary TES have several hurdles that impede their extensive implementation and effective functionality. These difficulties include technological, economic, and environmental dimensions, making confronting them for future progress essential. Critical challenges include augmenting energy storage efficiency, minimizing heat losses, optimizing system integration, and addressing constraints in thermal energy materials. Moreover, guaranteeing cost-effectiveness, scalability, and environmental sustainability is essential for TES to emerge as a viable alternative to traditional energy systems [50]. A primary technological difficulty is improving the energy storage efficiency of TES. Contemporary systems, especially those using PCMs, face challenges in efficiently storing and releasing thermal energy. Numerous materials have limited thermal conductivity, resulting in diminished heat transfer rates and decreased efficiency [55]. PCMs may absorb significant heat during phase transitions, yet their gradual energy release constrains their use in real-time scenarios. Researchers are investigating innovative materials such as nano-enhanced PCMs, which use nanoparticles to boost heat conductivity. Nonetheless, maintaining the stability and cost-efficiency of these materials for prolonged usage is a considerable difficulty [46].

2.3.1. Heat losses

A significant concern is thermal energy dissipation during storage and transfer operations. Heat loss transpires inherently owing to the temperature differential between the thermal energy storage and its environment, resulting in diminished system efficiency [56]. This may alleviate this issue; nevertheless, alternatives sometimes compromise efficacy and expense. The creation of cost-effective and eco-friendly

high-performance insulation is an urgent issue. Furthermore, preserving the thermal insulating characteristics of these materials over time, particularly under changing climatic circumstances, presents further hurdles. This is especially important for extensive thermal energy storage applications because minor inefficiencies might result in considerable energy losses [28].

2.3.2. System integration

Integrating with the current energy infrastructure is another significant hurdle for thermal energy storage. Although these systems provide viable energy storage and load-balancing alternatives, particularly renewable energy sources such as solar and wind, their integration into existing energy grids is intricate. Thermal energy storage systems must be able to store surplus energy when production surpasses demand and discharge it when demand is high [57]. Nevertheless, existing energy networks are not consistently designed to meet the particular requirements of thermal energy storage, including diverse demand patterns and variable temperatures. Furthermore, the control systems necessary for managing thermal energy storage integration sometimes lack the sophistication required for real-time modifications, particularly in dynamic situations like metropolitan regions or industrial settings. Smart grid technologies and sophisticated control algorithms are essential to address this difficulty; nevertheless, their deployment remains nascent [58].

2.3.3. Materials limitations

The materials used in thermal energy storage also pose considerable hurdles. Numerous materials demonstrating favourable thermal qualities, such as high heat capacity or phase transition capabilities, are either costly or possess restricted availability. This raises apprehensions over the economic feasibility of extensive thermal energy storage deployment, especially in cost-sensitive sectors such as construction and manufacturing [59]. Furthermore, some materials used in thermal energy storage, like synthetic PCMs or certain thermal insulation types, may exert detrimental environmental effects during their manufacturing processes or upon disposal at the conclusion of their life cycle. Pursuing sustainable, eco-friendly alternatives that maintain performance is a continuous scientific endeavour [60].

2.3.4. Economic challenges

Economic challenges are a significant impediment to the extensive use of thermal energy storage. The initial expenses related to installing and integrating these systems might be excessive, especially for smaller enterprises or household uses. Although thermal energy storage may result in substantial long-term energy savings, the requisite upfront expenditure often deters implementation [51]. Moreover, the payback time for thermal energy storage might be protracted, particularly in areas with low energy prices or minimal renewable energy subsidies. This hinders thermal energy storage's ability to compete with conventional energy storage or heating and cooling systems. Minimizing the expenses of critical components, including innovative materials and control systems, is crucial for enhancing the economic feasibility of thermal energy storage [50].

2.3.5. Environmental sustainability

Environmental sustainability is a serious issue in the development and application of thermal energy storage systems. These systems potentially offer reduced greenhouse gas emissions by promoting energy conservation and the use of renewable energy sources; however, their ultimate environmental impact is immensely affected by the properties of the materials employed and the assessment of their complete life cycle—from raw material extractions to syntheses, operational uses, and end-of-life disposal [52].

From this perspective, Life Cycle Assessment (LCA) acts as an essential tool for assessing the environmental impacts of nanomaterials in thermal systems. Lacking such an assessment, claims of sustainability

may prove to be incomplete or misleading. There have been many studies suggesting that even materials with very good thermal properties may generate high environmental costs through energy- and chemical-intensive manufacturing or due to problems on safe disposal and recyclability.

Toxicity of nanomaterials, especially carbon-based, and hybrid nanoparticles, remains poorly addressed in the literature. Such nanomaterials can potentially cause bioaccumulative or ecotoxic effects when released into the environment via leakage, degradation, or improper disposal. For instance, while multi-walled carbon nanotubes (MWCNTs) are extensively thermally conductive, studies have shown that they could induce oxidative stress and inflammatory response in biological systems upon certain exposure scenarios. A completely clear long-term picture regarding the behaviour of these nanoparticles concerning soil and water systems is still yet to be attained.

Thus, the development of nanofluids for TESSs should not be based exclusively upon thermal performance indices such as effective heat transfer or improvement of the Nusselt number. Rather, a balanced approach integrating environmental risk assessment, regulatory opportunities, LCA study, and toxicity investigation should be applied to provide assurance that the proposed solutions are effective in operation and soundly sensitive environmentally [53].

To further substantiate the sustainability discussion of nanomaterials used in TESSs, it is essential to consider quantitative metrics such as embodied energy, greenhouse gas emissions, and eco-toxicological effects [61]. For example, aluminium oxide (Al_2O_3) nanoparticles (widely used for enhancing thermal conductivity in NePCMs) have an estimated embodied energy ranging from 460 to 500 MJ/kg, with associated CO_2 emissions of 25 to 30 kg CO_2 -equivalent per kg produced. In contrast, copper oxide (CuO) nanoparticles, while offering higher thermal conductivity enhancement per unit mass, are associated with higher embodied energy (650–700 MJ/kg) and greater CO_2 emissions (35–40 kg CO_2 -eq/kg). These differences suggest that although CuO may deliver better thermal performance, its environmental manufacturing footprint is considerably larger than that of Al_2O_3 , raising concerns over large-scale deployment, especially in applications where sustainability is a priority [62].

Toxicological studies also indicate marked differences in ecological impact. Al_2O_3 nanoparticles typically exhibit low aquatic toxicity, with effective concentration (EC50) values generally above 100 mg/L, suggesting low risk to aquatic organisms. Conversely, CuO nanoparticles show moderate to high aquatic toxicity, with EC50 values often between 1 and 10 mg/L and are known to induce oxidative stress and membrane damage in freshwater and marine biota. Furthermore, while Al_2O_3 demonstrates good recyclability through thermal sintering and structural recovery, CuO suffers from degradation and oxidation challenges that complicate end-of-life handling. These observations underline the need for sustainability-aware nanomaterial selection, particularly in systems requiring high thermal performance alongside low environmental impact. Future research should emphasize green synthesis pathways, low-toxicity nanomaterials, and lifecycle-oriented design to minimize long-term ecological and human health risks [62].

3. Introduction to nanotechnology

Nanotechnology, derived from the Latin word *nanus*, meaning dwarf, is defined as the science and engineering dedicated to the design, synthesis, characterization, and application of materials and devices with at least one dimension on the nanometer scale (approximately 1–100 nm) or one billionth of a metre (10^{-9}) [63]. Nanotechnology is recognized as one of the twenty-first century's most innovative and cutting-edge technologies. It is an area of research constantly growing and contributes to practically every branch of science, including information and communications technology, medicine, materials science, agriculture, and the natural sciences and engineering. Researchers are particularly interested in these materials because their decreased size

may significantly affect their physico-chemical properties [64]. Numerous researchers have examined how adding nanoparticles affects the performance of composites made of polymers and fibres in ballistic applications, with encouraging findings [65,66]. Nanotechnology has made significant advances in water purification [67], agriculture [68], energy [69], transportation [70], electronics [71], healthcare [72], and environmental bioremediation [73]. Preclinical research has demonstrated the effectiveness of nanotechnology-based tools against various illnesses [74–76]. It has already demonstrated its value through its diagnostic, vaccine, and therapeutic applications, which have since expanded into clinical applications.

The treatment of COVID-19 has garnered significant interest due to the unique characteristics of nanoparticles [77,78]. Nanotechnology is becoming increasingly popular in the oil and gas [79]. The application of nanotechnology in the field of computer systems holds significant potential for enhancing the functionality and components of computers. Computing devices, on the other hand, have the potential to imitate and advance the development of nanotechnology [80,81]. New surface finishing and production processes are now being considered by researchers as a potential means of using nanotechnology to develop applications that are both novel and intelligent [82]. The packaging of food, which was traditionally passive or conventional, is currently transitioning toward active or creative packaging that makes use of nanotechnology to develop food packaging that is adaptive, dynamic, and intelligent, with increased functionality [83,84]. Before implementing nanotechnology into everyday life, it is necessary to take into consideration a number of potential concerns, despite the fact that it offers a multitude of benefits. Nanoparticles are very different from their bulk counterparts in terms of the characteristics that they show. As a result of decreased size and increased surface area per unit volume, migration of nanomaterials into food from packaging, and exposure of consumers to nanoparticles, there is a possibility that potentially harmful effects might occur on human health [85,86].

3.1. Nanomaterials and their properties

Any substance with at least one dimension inside the nanoscale range (generally between 1 and 100 nm) is called a “nanomaterial”. Examples of the various structures classified as nanomaterials include thin films, nanoparticles, nanofibers, and nanotubes. Nanomaterials’ huge surface area and compact size make them attractive because they increase the ratio of surface atoms to inner atoms. As a result, the surface chemistries of bigger materials that are reduced to nanoscale dimensions become more prominent and alter their physical characteristics [87]. Metals, polymers, natural chemicals, and nanostructured materials are the four primary categories into which nanomaterials can be classified [88]. Copper [89], Gold [90], palladium [91], selenium [92] and are examples of metals, which are typically mineral nanoparticles.

Six conventional methods are primarily used to produce nanomaterials:

- A solid is broken up into nanocrystallites by a process known as ball milling [93]. This process involves letting very small balls that circle a drum to fall on the drum due to the force of gravity. By utilizing this method, it is possible to generate fines as well as dispersions of homogeneous oxide particles.
- Sol-gel synthesis [94]: The technique of creating networks by first creating a colloidal suspension (sol) and then gelating the sol to create a network in a continuous liquid phase (gel) is known as sol-gel synthesis. The ions of a metal or metalloid element encircled by various reactive ligands create the colloids.
- Electrodeposition [95]: Controlled application of one or more layers to a surface is known as electrodeposition. Particles are placed on the graphene sheet’s surface due to the electrodeposition process.
- Plasma arcing [96]: Gas is converted to conduct electricity during plasma arcing, finally releasing its electrons and ionizing. Plasma is

the term for the ionized gas that results. Carbon nanotubes can be formed using this technique. Among the forms of plasma discharge are vacuum arcs.

- Chemical vapour deposition [97]: The process of chemical vapour deposition involves the creation of nanoparticulate particles from gas. After the material has been heated to the point where it becomes a gas, it can frequently settle as a solid on a surface while being vacuumed. It is possible to create a new product that is separate from the material that has been volatilised by either direct deposition or a chemical reaction.
- Using natural nanoparticles [98]: To employ natural nanoparticles, their crystallite ties must be broken to be reconstructed into new materials. Batteries, motor vehicles, aircraft, medical implants, machine tools, insulating materials, and high-power magnets are all made with nanomaterials.

3.2. Nano-enhanced phase change materials

The storage of thermal energy is essential for the implementation of renewable energy on a large scale as well as the transition to decarbonised buildings and transportation networks. Latent heat storage is included in the category of thermal energy storage. The most beneficial approach allows for the storage of solar thermal energy through the use of PCMs latent heat. Thermal energy storage, heat transfer rate, and thermal conductivity are all improved when nanoparticles are added to PCM. PCM has a variety of applications, including but not limited to: a) electrical devices; b) solar thermal systems [99], c) building thermal management; d) cold storage for food and medication; and e) photovoltaic thermal panels. As soon as the temperature of the surrounding environment drops below the phase transition temperature of the PCM, the thermal energy that has been absorbed is released into the environment [100]. The term “enthalpy of fusion” refers to the thermal energy that is either absorbed or released during the phase transition of PCM from solid to liquid and liquid to solid modes [101,102]. During the phase transition, the PCM either takes in or gives off a significant quantity of heat, which is referred to as latent heat. This occurs without the PCM creating a change in temperature. The addition of a source term to the equation that represents the latent heat allows for this to be incorporated into the equation that describes heat conduction.

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} + \frac{1}{\rho c_p} \frac{\partial H}{\partial t} \quad (1)$$

where c_p is the specific heat capacity, ρ is the material density, $\frac{\partial H}{\partial t}$ is the rate of latent heat absorption or release.

The quantity of heat that PCM stores or releases during a phase transition can be explained as follows:

$$Q = mL \quad (2)$$

In this equation, Q represents the amount of heat absorption or release, m represents the mass of the PCM, and L represents the latent heat of fusion or solidification.

The latent heat from phase change and the sensible heat from temperature change must be considered when determining the total energy released or stored. The following is an expression for the total heat (Q_{total}):

$$Q_{total} = mc_p \Delta T + mL \quad (3)$$

where ΔT is the change in temperature during the cooling or heating process or after phase change occurs.

PCM heat storage and transmission may be simulated using this mathematical statement, which provides a fundamental basis for the simulation. By solving these equations, it is possible to make predictions regarding the temperature distribution as well as the amount of thermal energy that the PCM stores or releases. This is a key step in the process of

developing effective thermal management and energy storage systems. The categorisation of materials that undergo phase change is shown in Table 2.

The addition of nanoparticles to PCMs offers comprehensive solutions to the inherent limitations of conventional PCMs, such as low thermal conductivity and phase separation. This innovation has led to the development of NePCMs, which incorporate nanoparticles (ranging from simple 0D structures like Al_2O_3 and CuO to complex 3D architectures) into the PCM matrix. These nanoparticles significantly enhance thermal conductivity and heat transfer performance. For example, 0D nanoparticles, such as metal oxides, improve thermal diffusion, making NePCMs ideal for rapid thermal applications. The degree of enhancement depends on particle characteristics including size, shape, concentration, and dispersion stability. Throughout this study, the term NePCM will consistently refer to these nano-modified PCMs. 0D nanoparticles, such as quantum dots or nano-spheres, scatter electrons, and phonons to increase the thermal conductivity of PCMs. 0D-enhanced PCMs are perfect for rapid heat applications. Ultrafine nanometer-scale particles with diameters under 100 nm larger than atomic clusters are called 0D nanoparticles. Metal and its oxide nanoparticles, including ZnO, Cu, Fe, Ag, Al_2O_3 , and TiO_2 .

The study by Rufuss DDW et al. [105] concentrated on creating NePCM by utilizing titania and copper oxide nanoparticles in paraffin. It was found that adding 3 % titania to the base PCM increased the value of k by 25 %, whereas adding copper oxide nanoparticles increased the value to 28.8 %. Praveen and Suresh [106] experimented by applying CuO nanoparticles in Neopentyl Glycol. Using PCM resulted in a significant enhancement of Thermal conductivity by a factor of 4. One-dimensional nanoparticles include nanorods, nanotubes, and nanofibers, among other examples. One-dimensional materials include carbon nanotubes and additions of 1D metal compounds such as copper sulphide, zinc oxide, and molybdenum disulphide. Crystalline structure is responsible for the excellent performance of carbon fibres, which is measured in terms of the k value, which is around 1000 W/(m·K).

The research conducted by Li et al. [107] involved the formation of

carbon nanofibers into flake graphite. This resulted in the creation of a hybrid structured nanomaterial that enhanced the thermal performance of PCM. Within the context of the formation of stable PCM composites, the findings demonstrate an increase in thermal conductivity, loading capacity, and overall thermal efficiency. In their research, Cao et al. [108] synthesised NePCM by functionalising Single-Walled Carbon Nanotubes (SWCNTs) and Multi-Walled Carbon Nanotubes (MWCNTs) using hexadecyl acrylate that was used in the preparation process. At 0.47 W/(mK) and 0.88 W/(mK), respectively, the thermal conductivity values of SWCNTs and MWCNTs were 134 % and 339 % higher than the basic PCM. This comparison was made using the basic PCM as a reference. The use of nanoparticles appears to have resulted in a significant enhancement in the thermal performance of the basic PCM. Examples of two-dimensional nanomaterials include graphene and its derivatives, MXene, molybdenum disulphide (MoS_2), and hexagonal boron nitride (h-BN) [109]. By interacting with PCM molecules, graphene and its derivatives are able to obtain considerable gains in heat conductivity. This is because of the vast surface area that these molecules possess. Aslfattahi [110] and colleagues dispersed MXene with organic PCM (paraffin) in order to improve the thermo-physical properties of the compound.

The intermolecular interactions that take place between the atoms in the surface area of the MXene nanoparticles and the C—H bonds of the organic PCM are responsible for the enhancement of the light-to-heat energy conversion procedure. As part of their research, Guo et al. [111] investigated the thermal energy storage and heat dissipation capabilities of the PCM with fin metal foam hybrid assembly. When compared to the baseline scenario, the use of fin metal foams in the thermal energy storage unit resulted in an improvement in temperature uniformity, an increase in the transient temperature response, and a reduction in melting time that was 83.35 % shorter. It is possible to make nano-enhanced PCM by spreading nanoparticles throughout the matrix of the PCM. The NePCMs have garnered a lot of attention because to the fact that they have the potential to be utilised in a variety of fields. The thermo-physical characteristics of nano-enhanced PCM and PCM are clearly differentiated from one another as a result of the presence of nanoparticles. The incorporation of nanoparticles has the potential to influence the melting temperature of the PCMs in a substantial way. Both the PCM matrix and the nanoparticles interact with one another, which has an effect on this transition happening.

The addition of nanoparticles, particularly those with a high thermal conductivity, such as graphene or carbon nanotubes, results in a considerable increase in the thermal conductivity of PCMs. There are a number of factors that influence this enhancement, including the concentration, size, shape, and distribution of the nanoparticles that are included inside the PCM. The presence of nanoparticles has an effect on the density of PCMs, which is an essential factor in determining the volume consumption requirements of thermal energy storage devices. The results of the introduction of nanoparticles by a number of researchers are shown in Table 3, along with the qualities that are altered as a result. The findings of a number of experiments point to the likelihood that nanoparticles might significantly increase the performance of PCM. These advancements not only make PCMs more efficient in terms of controlling and storing thermal energy, but they also expand the spectrum of applications that they may be used for.

3.3. Stability issues: agglomeration, phase separation, and cycling durability

Despite the promising enhancements in thermal conductivity and heat transfer performance, NePCMs suffer from critical stability challenges, particularly nanoparticle agglomeration and phase separation. These issues compromise the uniformity of dispersion, reduce the effective surface area for heat transfer, and lead to irreversible deterioration in thermal energy storage properties after multiple melting-solidification cycles [122].

Table 2
Classification Scheme for PCMs.

Based on the Physical state	Organic	Paraffin, Ester, Fatty Acids, Alkanes, Alcohols
	Inorganic	Salts, Metals, Metal Alloys
Based on the melting temperature of the material	Low temperature	phase change lower than room temperature. These are suitable for refrigeration and cold storage.
	Medium temperature	Phase change temperature is between room temperature and 100 °C. Electrical cooling and construction materials use PCMs.
	High temperature	Applications necessitating high-temperature heat storage are made with phase change temperatures exceeding 100 °C. Molten ions and metal alloys are two examples of high-temperature PCMs.
	paraffin	Thermal energy storage often uses organic PCM. Hydrocarbon paraffins are mostly carbon and hydrogen. Octadecane and eicosane are paraffins [103].
Based on the chemical composition of material	easter	organic PCM is a hydrocarbon chain having a —COOH end. Saturated or unsaturated fatty acids have single or double bonds. Fatty acids are eco-friendly and biodegradable.
	salt hydrates	Inorganic PCMs are salt-water mixtures. Calcium chloride hexahydrate, magnesium sulphate heptahydrate, and sodium sulphate decahydrate are common [104].

Table 3

The effects of the properties of nanoparticles.

Properties	PCM	Nanoparticles	Observation
Melting point	sodium acetate trihydrate (CH ₃ COONa·3H ₂ O, SAT) [102]	copper	The PCM melting temperature was reduced to 57.8 from 58.5 °C.
	SAT-acetamide [112]	aluminium nitride	The 9 wt% addition of AlN nanoparticles reduced the composite melting temperature from 47.8 to 47.3 °C
Thermal conductivity	Paraffin [113]	carbon nanotubes (SWCNT)	The thermal conductivity of PCM is enhanced by factors such as the particles' weight fraction, size, and form.
	Paraffin [114]	titanium oxide (TiO ₂)	Up to a certain point, the heat conductivity gets better. However, the thermal conductivity falls when the temperature rises over this point, surpassing the PCM melting temperature.
Latent heat capacity	Paraffin [115]	Al ₂ O ₃ and Fe ₂ O ₃	In the resulting NePCM, the latent heat capacity gradually decreases. The main explanation could be a decrease in the PCM's mass fraction.
Thermal diffusivity	polyethylene glycol (PEG 4000) [116]	GNP	there is a steady drop in latent heat
	eutectic PCM of lauric acid and methyl palmitate [117]	GNP nanoparticles	The results exhibited an improvement in thermal diffusivity.
	RT50 PCM [118]	graphene	The results demonstrated the optimum 34 % increment in thermal diffusivity
Density	eutectic (palmitic-paraffin) [119]	titanium oxide	a decrement in thermal diffusivity is observed
	paraffin [120]	graphene	The experimental outcomes revealed a noticeable rise in the density of the NEPCM, from an initial value of 800 kg/m ³ to a final value of 923 kg/m ³
Specific heat	eutectic solution, including lithium carbonate and potassium carbonate [121]	nanoplatelets (GNP)	a significant enhancement in the specific heat capacity of the resultant nanofluid was attained, with gains ranging from 19 % to 24 %

Nanoparticle agglomeration occurs due to van der Waals attractions and insufficient surface functionalization, resulting in clusters that act as thermal barriers. Phase separation, on the other hand, arises from density mismatches between the base PCM and embedded nanoparticles, especially under the influence of gravitational and thermal cycling forces. Both mechanisms deteriorate the homogeneity of the composite and negatively influence the latent heat capacity and melting point.

Several experimental investigations have highlighted this concern. For instance, researchers observed a 12 % reduction in latent heat capacity after 200 thermal cycles in paraffin-based NePCMs with CuO nanoparticles due to agglomeration. Similarly, different work reported phase separation-induced sedimentation in Al₂O₃-enhanced PCMs, leading to localized overheating and reduced heat storage efficiency [123]. To mitigate these effects, researchers have proposed:

- Surface modification and functionalization of nanoparticles (e.g., with surfactants, silanes, or polymer coatings) to improve colloidal stability.
- Use of supporting matrices such as porous carbon, metal foams, or encapsulation shells to physically stabilize the nanoparticles within the PCM.
- Ultrasonic dispersion and high-shear mixing during preparation to ensure better distribution.
- Incorporation of thixotropic agents or stabilizers, which enhance the viscosity and reduce sedimentation.

In recent work, carbon-based nanostructures such as graphene oxide and CNTs have shown superior stability due to their high aspect ratio and strong interfacial bonding with the PCM matrix, sustaining performance over more than 500 cycles without notable degradation [122].

Therefore, assessing the long-term cycling behaviour and implementing stabilization strategies are essential prerequisites before NePCMs can be considered for commercial-scale deployment in thermal energy systems.

3.4. Nanofluids for enhanced heat transfer

It is possible to transform a fluid into a nano-fluid by incorporating nanoparticles into the mixture. Nanofluids are liquid suspensions of nanoparticles that are dilute and the diameters of at least one of the nanoparticles are smaller than 100 nm [122]. The components that make up nanofluids are a base fluid and a dispersion of very small particles. There are a variety of possible base fluids, including water, ethylene glycol, surfactant-based fluids, polymeric fluids, engine oil, and

mixtures of two or more liquids, among others. Nanofluid research is an interdisciplinary field that draws from the fields of physics, chemistry, and engineering. It has garnered a significant amount of attention because it has the potential to be applied in a variety of different industries. Choi and Eastman [123] presented the idea of nanofluids and started conducting experiments with the incorporation of nanoparticles.

Research on viscosity, stability, and other features that are essential for practical applications has been added to the investigations that were initially focused on improving thermal conductivity. Nanoparticles are combined with base fluids to produce a wide variety of nanofluids. Some examples of these nanofluids are copper oxide (CuO), alumina (Al₂O₃), silver (Ag), zinc oxide (ZnO), titanium dioxide (TiO₂), carbon nanotubes (CNT), iron oxide (Fe₂O₃), magnesium oxide (MgO), cerium oxide (CeO₂), and metal nanoparticles (Au). Bio-fluids, ethylene glycol, water, and a variety of coolants and lubricants are examples of basic fluids that are commonly used. The most recent advancements in nanofluids and the mathematical modelling of these fluids play an important part in the field of nanotechnology and industrial applications. However, the poor thermal conductivity rate of ordinary base fluids, such as water, ethylene glycol, and oil, is the fundamental restriction. In today's world, the cooling of electronic equipment is the most important industrial necessity due to the rapid advancement of technology [124,125]. The nanoscale solid particles are submerged into host fluids in order to circumvent these constraints. This process alters the thermophysical properties of the host fluids and significantly increases the rate at which heat is transferred [126–129].

Nanofluids find their uses in a wide variety of fields, including but not limited to the following: cooling of electronic devices, heat exchangers, nuclear reactor safety, hyperthermia, biomedicine, engine cooling, car thermal management, and many more [130–133]. A variety of mathematical models have been utilised by a number of writers in order to provide a description of heat transport in nanofluids. Out of all of these models, the ones that are utilised the most are the ones in which the concentration of the nanoparticles remains constant, and the incorporation of the nanoparticles into the base fluid results in an improvement in the nanoparticles' physical attributes [134–136]. In addition, there are other models that include thermal dispersion [137] and Brownian motion [138] that are based on the modification of the physical parameters. The thermal properties of base fluids have been investigated by Buongiorno [139], who has devised a mathematical nanofluid model that is more complex than the one previously used.

Brownian motion and thermophoresis are two of the most important mechanisms for the transport of particles in nanofluids because of their similarities. Nanoparticles in the fluid have a tendency to be uniform

when the Brownian motion force is applied. It is the temperature gradient in the base fluid that is the source of the thermophoresis force. As a result of the fact that the particle receives more momentum impacts from the fluid molecules on the hot side than that on the cold side, the particle has a tendency to move in a direction that is opposite to the temperature gradient (the particle moves from hot to cold). This occurs when the size of the particle is very fine (on the order of nanometres). In light of this, nanofluids possess mass transfer processes as a result of the Brownian motion and thermophoresis forces. Energy is carried by the nanoparticles that migrate, and they transfer that energy to the medium that surrounds them.

The flow of nanofluid was tested by the researchers under a variety of settings and with a variety of nanoparticles. With the help of Buongiorno's and non-homogeneous models, Celli [140] investigated the free convection flow that occurred within a side-heated square cavity that was filled with a nanofluid. There is a presumption that the thermophysical parameters of the nanofluid are functions of the average volume fraction of nanoparticles that are spread within the cavity. In another study [141], Sheikholeslami and colleagues investigated the MHD natural convection of a Cu–water nanofluid that was contained within a square cavity that was heated from below and cooled from the side walls. It has been found that the rate of heat transfer experiences a drop with the Hartmann number, while it experiences an increase with the length of the heat source, the percentage of nanoparticle volume, and the Rayleigh number.

Zhang et al. [142] and Shahsavar et al. [143] conducted an experimental investigation on the thermal conductivity of nanofluids by utilizing a variety of base fluids. The researchers investigated the ways in which the concentration of the dispersant, along with other parameters including mass fraction, temperature, and standing time, altered the thermal conductivity of the CNT-nanofluid blend. Following their discussion, they made the observation that the temperature range for 20–60 °C is 0–7 %, and the volume percentage range is 0–7 %. It was observed that the thermal conductivity of the nanofluid increased as both the temperature and the volume % rose. Nanofluid has been the subject of extensive study that has been conducted to investigate its potential as an alternative coolant for electronic cooling systems. These studies have demonstrated that nanofluid has the ability to increase the efficiency of heat transmission in microchannel heat sinks [144].

The researchers Lau et al. [145] discovered that nanofluids that contained alumina particles had a maximum temperature reduction of 7 °C when compared to regular water. Sajid and Ali [146] present a review of the application of nanofluid in the cooling of a variety of thermal devices, which has been the subject of research by a large number of researchers and practitioners. Nanofluids are utilised by Olabi et al. [147] in order to improve heat transmission in a variety of heat exchanger configurations. We investigated the heat transport of the conjugate in a container that was filled. Table 4 presents the results of a number of recent investigations that were conducted on nanofluids contained within a variety of enclosures. In general, the findings of these experiments demonstrated that the incorporation of nanoparticles into conventional working fluids leads to a notable enhancement in the flow and heat transfer characteristics of the fluids.

To provide a more transparent comparison of the thermal performance improvements achieved using various nanomaterials in PCMs and base fluids, Table 5 summarizes key findings from selected studies. The table includes data on thermal conductivity, specific heat, and latent heat capacity enhancements as a function of nanoparticle type, concentration, and PCM/base fluid. These results highlight the significant variability in thermal performance depending on nanoparticle selection and concentration, demonstrating the importance of material optimization for targeted thermal energy applications.

4. Artificial intelligence in thermal energy systems

This section provides an overview of AI in TESSs.

Table 4

Recent studies of nanofluid inside various enclosures.

Author	nanofluids	configuration	Observation
Dulami et al. [148]	Ag–MgO/water hybrid nanofluid	porous corrugated enclosure	thermal transfer performance is greatly improved as the volume percentage of nanoparticles increases
Al-Srayyih et al. [149]	Cu–Al ₂ O ₃ /water hybrid nanofluid	quarter-circle having a pair of oscillating elliptical obstacles	transfer of heat increases with nanoparticle addition
Parveen et al. [150]	Fe ₃ O ₄ /water	Inclined wavy cavity	the increase in nanoparticle volume fraction increases the effective electrical conductivity. Their findings demonstrated that increasing the nanoparticle volume fraction enhanced the heat transmission rate.
Al-Amir et al. [151]	TiO ₂ -water nanofluid	Z-Staggered cavity	The addition of nanoparticles influences the convective effect. A 6-vol% CuO nanofluid composition exhibited an 89 % increase in heat transfer rate compared to the base fluid.
Rehman et al. [152]	Cu–Al ₂ O ₃ /water	Inclined wavy cavity	the Nusselt number increases with an increase in nanoparticle volume fraction
Vajjha et al. [153]	CuO/Ethylene glycol and water mixture	Automotive radiator flat tube	nanoparticle volume fraction enhances the heat transfer rate
Faraz et al. [154]	Cu-water	hexagonal enclosure fitted with a square cavity	Percentage analysis indicates that hybrid nanofluids (Fe ₃ O ₄ – MWCNT) significantly improve thermal distribution compared to conventional fluids.
Rahimpour and Moraveji [155]	Fe ₃ O ₄ /water	C-shaped	Changes in the volume percentage of nanoparticles upgraded the heat transfer rate inside the enclosure.
Qurrat Ul Ain et al. [156]	Fe ₃ O ₄ -MWCNT	Star-shaped	
Hansda [157]	Fe ₃ O ₄ -Cu/water hybrid	Wavy trapezoidal	

4.1. Brief overview of AI in energy management

AI and ML are emerging energy concepts that can enhance system effectiveness by incorporating past and predicted futures [158]. The “Fourth” Industrial Revolution heavily relies on AI, which is expanding quickly and having a big impact on the world economy. AI is a multi-disciplinary field that creates intelligent computer programs using data-driven methodologies and sophisticated algorithms [159,160]. Though AI has been used extensively in the energy industry recently, it has also been used in a number of other sectors, such as finance, healthcare, education, and homes. AI was first applied to basic research and experimental tasks like power system prediction, fault diagnosis, and optimization. However, AI hasn't been used extensively in the energy industry because of gaps in data collection and processing power. AI also necessitates IT resources exclusive to computing centres, which have a big environmental impact and substantial energy consumption [161].

AI is applied in different aspects of energy. Energy usage modelling for buildings is one application of AI and ML [162]. The task of determining a building's energy use is complicated because of several variables, including equipment, occupant behaviour, weather, and physical attributes. Energy usage has been modelled in research using techniques

Table 5
Comparative performance metrics of nanofluids and nano-enhanced PCMs from selected studies.

Study	Nanoparticle	Base Material	Concentration (wt %)	Thermal Conductivity Enhancement (%)	Specific Heat Change (%)	Latent Heat Change (%)
Rufuss et al. [105]	TiO ₂	Paraffin	3	25	–	–
Praveen & Suresh [106]	CuO	Neopentyl Glycol	2	400	–	–
Cao et al. [108]	MWCNT	Hexadecyl Acrylate	0.47	339	–	↓12
Zhang et al. [142]	CNT	Water	0.5	27	+8	N/A
Shahsavari et al. [143]	GNP	PEG 4000	1	35	–6	↓10

like support vector machines, decision trees, and ANN. This field also uses genetic algorithm and ANN techniques to optimize thermal comfort control in HVAC systems and absorption chillers. In the chemical industry, AI and ML are promising for process control applications [163,164]. Secondly, utilizing AI and ML in renewable energy is becoming more common, especially in photovoltaic power generation. These techniques can predict important variables and assist designers in improving installation site setups by forecasting and optimizing power generation. This can lead to a significant improvement in power generation. A variety of techniques can be applied to forecast power generation, such as ANN, historical data, and weather prediction.

AI and ML techniques can also be used to model wind energy, another renewable energy source. Additionally, ML has the potential to control and supply renewable energy for electric car batteries. Researchers are becoming interested in new AI and ML-based models, such

as ANN and fuzzy logic models, which show bright futures, while physical and conventional static models are effective for long-term energy generation predictions [164–166]. AI and ML are also being applied to energy storage to use energy sources if they are unavailable effectively. The architecture, performance, and characteristics of Li-ion batteries have all been described using ML techniques, which have also been used to suggest new materials that could increase stored energy. A recent popular idea is the Internet of Things (IoT), which connects commonplace items like buildings, appliances, cars, and people to the Internet and networks. By maximizing the flow of electricity through the grid by lowering power drawn and raising the power supply, IoT systems seek to minimize energy consumption, expenses, and emissions [163,167].

Energy systems can monitor and manage energy production, consumption, and storage while cutting costs and balancing supply and

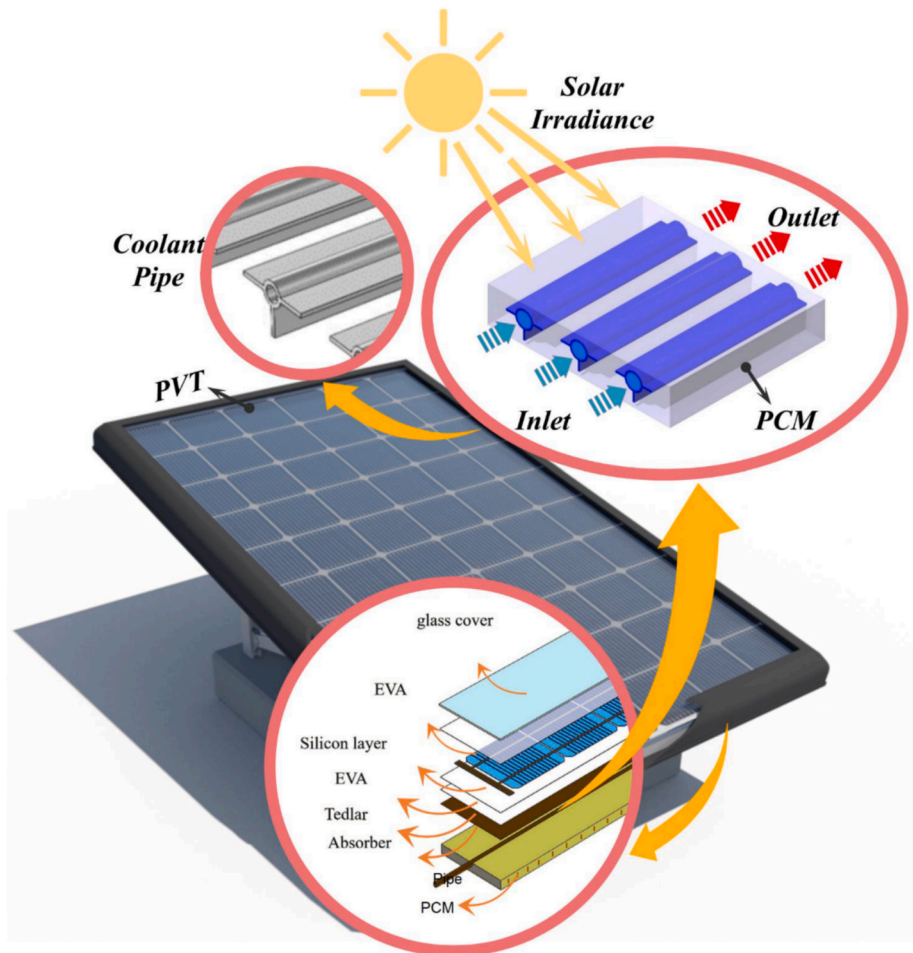


Fig. 2. AI-optimized solar collector using nanofluid and PCM for improved cooling and efficiency. The schematic illustrates the integration of AI algorithms like predictive control with nanofluid flow channels and PCM modules.

demand. By imitating natural processes like swarm intelligence, genetic evolution, and animal cognition, AI-based metaheuristic optimization technique solve challenging energy management problems and are dependable and effective energy management systems [168]. The metaheuristic optimization technique use three different search types to find optimal solutions: global, local, and random [169]. To optimize the energy system, these algorithms examine enormous amounts of data, spot complex patterns, and make deft choices. Drawing inspiration from the various ways that metaheuristic optimization techniques contribute to the energy system, this study concentrates on the following methods: Genetic Algorithms (GA) [170], PSO [171], ant colony optimization (ACO), simulated annealing (SA), Cuckoo Search Algorithm (CSA), and a few more recent ones [168].

In Fig. 2, a flat plate collector is mounted on top of a PCM module, which has channels to circulate the flow. These channels allow the heat that is transferred to the nanofluid that flows into them through the solar panel and PCM. AI and ML are used in the given system to identify the conditions under which it should work to be optimal. These are choosing the right diameter of pipes, changing the flow rate of nanofluids, and the contact with PCM. The main objectives include the prevention of high operating temperatures of the solar panel, maximizing heat extraction ability into the nanofluid, and having a minimal pressure drop throughout the circulation circle. This intelligent control enables better thermal management and improved system performance.

Fig. 3 reveals a detailed schematic workflow that demonstrates how the design and optimization of heat exchangers can be integrated with AI. The concept of the core of this framework is a hybrid AI approach using GA and ANNs, which provides a valuable mechanism of automatic and intelligent design exploration. The role of the GA in this system is that of the global optimizer that creates a multitude of possible design variants by controlling important parameters like tube diameter, fin spacing, the rates of fluid flow, and geometrical patterns. The trained ANN model is then used as a surrogate or meta-model to predict the thermal and hydraulic performance metrics, i.e. Nusselt number, pressure drop, and overall heat transfer coefficient, using the input design characteristics of these design variants. Such hybrid GAANN system dramatically eliminates the requirement of computationally expensive simulations or experiments at every step. Rather, ANN model quickly approximates performance outcome, making it possible to converge within optimum designs more quickly.

4.2. AI algorithms and techniques

Finding applications for AI algorithms in the energy sector requires understanding this vital area of computer science. These involve various scientific concepts and technologies, with computer systems drawing

conclusions based on human logic and intelligence. This section presents an overview of some of these algorithms and techniques.

4.2.1. Machine learning

Due to renewables and EVs, power grids are evolving into smart Energy and Electric Power Systems (EPPS), incorporating integrated Cyber-Physical-Social Systems (CPSS). These systems generate high-dimensional data resources, generating large volumes of data. AI and ML tools can efficiently handle the difficulties of this high-dimensional, nonlinear data, time-varying, and the impacts of external and internal multi-physical parameters in EPPS [172]. In 1959, Arthur [173] introduced the concept of ML, which allows computers to learn without explicit programming. ML aims to identify patterns in data and use those patterns to forecast results. This allows computers to explore and discover huge volumes of data, which helps humans make decisions. The goal of ML is to find an objective function f that creates the best possible mapping between an input variable (X) and an output variable (Y), allowing for more precise output predictions and evaluations $Y = f(X)$ [172]. Fig. 4 shows the various approaches of ML.

The kind of input data and its quality impact how well ML algorithms work. ML is more than just a set of techniques, algorithms, or prediction models; instead, several classifications exist in the literature. Effective ML strategies can be found by synthesizing the literature according to algorithm type and input data. The following are some of the techniques of ML [174,175]; RL, Supervised Learning (SL), Semi-Supervised Learning (SSL) and unsupervised learning.

4.2.1.1. Reinforcement learning. With roots in statistics, control theory, and psychology, RL technology became widely studied and used in the late 1980s and early 1990s in domains such as automatic control, ML, and AI. This significant ML technique has three varieties: non-supervised learning, RL, and SL. Using the reinforcement signal from the environment, RL is an online learning technology that evaluates an intelligent agent's action quality without giving it instructions on how to generate the appropriate action. To learn and adjust to the environment's appraisal value, the intelligent agent depends on experience because the external environment offers little information [176]. RL is a technique that maximizes a numerical reward signal by mapping situations to actions. It assumes that a computer system is given a set of rules and then observes the results by those rules. Dynamical systems theory, in particular optimal control of Markov decision processes, can be used to formalize RL problems [177].

4.2.1.2. Supervised learning. During training, SL techniques need labelled data and utilize pre-established patterns and examples. Because it uses labelled data and can map or estimate the results of a fresh

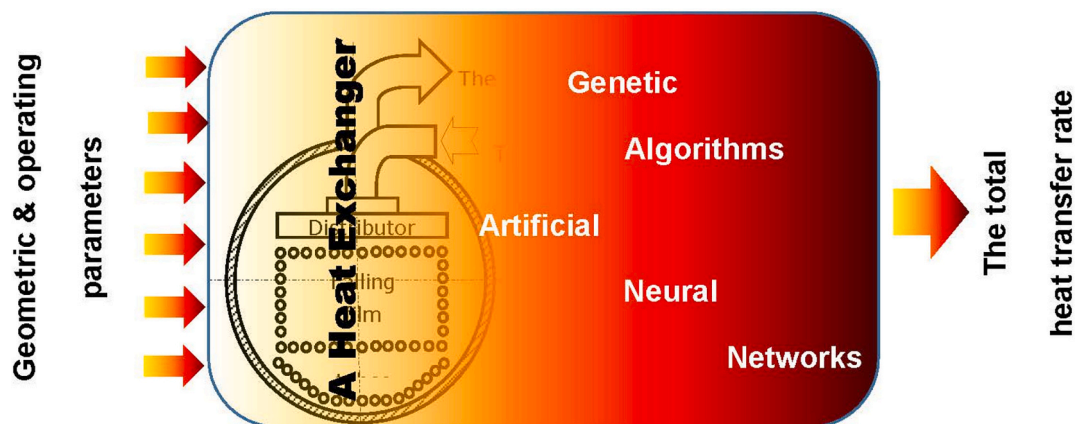


Fig. 3. AI-Driven optimization workflow for heat exchanger design combining GA and ANNs. The diagram outlines the hybrid AI approach, where GA generate design variants, and ANNs predict thermal performance metrics like Nusselt number.

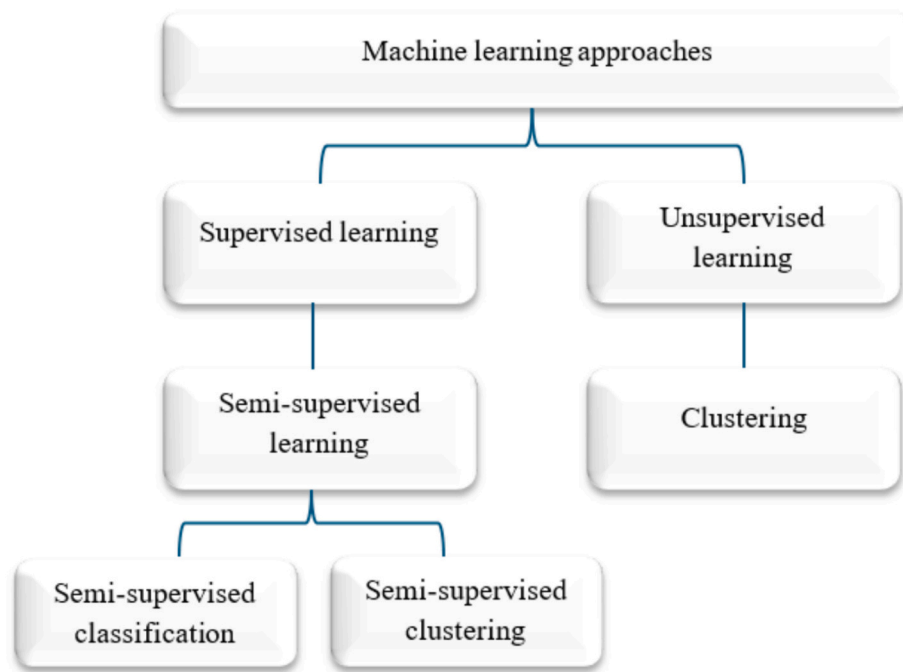


Fig. 4. Approaches of ML.

sample, this approach is more accurate than unsupervised learning. The time and specialized knowledge needed to label the data, which is a major challenge, are the drawbacks, though [177–179]. Regression problems are the main application for SL algorithms, including decision trees, Random Forests (RFs), logistic and linear regression, support vector machines, ANN, and other neural network algorithms [180].

4.2.1.3. Semi-supervised learning. A ML technique called SSL was developed to tackle the shortcomings of both supervised and unsupervised learning. Large amounts of training data are necessary for SL, but unsupervised learning cannot reliably cluster unknown data and does not require any labelled data. Researchers suggest SSL, which can identify unknown or test data and gain knowledge from a limited quantity of training data, to solve these problems. When building a model, SSL uses a few labelled patterns as training data and the rest as test data. SSL is further separated into two categories: semi-supervised clustering and semi-supervised classification [181].

4.2.1.4. Unsupervised learning. Data points are frequently grouped or clustered using unsupervised learning algorithms, which analyse a dataset by looking at shared characteristics. Additionally, they summarize and explain data features like hierarchical clustering, principal component analysis, and k-means clustering. Additionally, there is semi-unsupervised learning, in which only a small portion of the training dataset has labels [180]. Optimisation algorithms and principal component analysis are two examples of the many diverse paradigms and characteristics that are associated with AI algorithms. Through dimensionality reduction, the number of features that are taken into consideration is reduced. This is accomplished by concentrating on the characteristics that are responsible for the majority of the variance in the output variable. For the purpose of locating a workable solution for a function, optimisation algorithms, such as gradient descent, grey wolf optimiser, and GA, make use of stochastic and heuristic principles. Examples of ML domains include natural language processing, which analyses and forecasts text and graph-based algorithms, and neural networks, which convert data into network graphs for analysis based on centrality and similarity. Both of these domains are associated with the field of ML [182].

4.2.2. Deep learning

DL, a subset of ML, reduces the need for manual feature extraction by modelling complex data patterns using multi-layered neural networks. This makes it useful for tasks like natural language processing, image and speech recognition, and gaming [183]. The Fourth Industrial Revolution (Industry 4.0) focuses on automation and smart systems in various applications like cybersecurity, smart cities, business intelligence, and smart healthcare, which are driven by technology. Due to its ability to reveal intricate architecture in high-dimensional data, DL has greatly enhanced performance in various applications, most notably security technologies. Due to their ability to learn from past data, DL techniques can be extremely important in developing intelligent data-driven systems. Through automation and experience-based learning, DL has the potential to transform the world and the way people live their daily lives. AI, ML, and data science with advanced analytics, especially in intelligent computing, are all impacted by DL technology. This conversation will examine DL's role in AI and how it relates to these fields [184,185]. In power engineering applications, DL algorithms are mostly divided into three major classes according to theoretical considerations, discriminative deep ANN, probabilistic deep neural networks ANNs and Deep Reinforcement Learning (DRL) [186].

4.2.2.1. Discriminative deep ANN. Learnt using discriminative deep ANNs are highly nonlinear decision boundaries between different classes and regression regions of power system data. Online small signal stability evaluation, faulty line localisation, and the rectified linear unit ANN for real-time responsive reliability management are all part of this. The stack autoencoder was developed for the purpose of unsupervised pattern identification in the areas of fault diagnostics, PV power forecasting, and wind energy prediction. The sequential behaviour of the time-dependent power system data is represented by a supervised temporal feature extractor called a long short-term memory ANN. By utilizing convolutional and pooling operations to capture persistent patterns in power system measurements, convolutional neural networks accomplish spatiotemporal tasks in areas such as fault identification, transient stability analysis, and renewable energy forecasting.

4.2.2.2. Probabilistic deep neural networks ANNs. To determine which

hidden variables best characterize the data's probability density function, probabilistic deep neural networks ANNs employ feature learning. The popular probabilistic graphical model known as deep belief networks uses conditionally independent latent features to learn the probability density function of data. They are employed for probabilistic state estimation, load prediction, transient stability evaluation, and wind and solar power prediction. Generative Adversarial Networks (GANs) are models that compare their generated data samples with the real dataset to improve the accuracy of their learned probability density function. They are applied to model-free renewable scenario generation, smart grid cyber-attack detection, and outlier and fault detection issues. To determine the probability density function of data, variational autoencoders (VAEs), a new type of deep generative ANN, learn a high-dimensional latent variable that is mapped to the original data samples. Electric vehicle load generation, power grid synthesis, and unsupervised anomaly detection in energy time series are all applications for VAEs.

4.2.2.3. Deep reinforcement learning. DRL is an ML method that makes use of environmental feedback, calculated by a reward function, to learn the best course of action. In contrast, traditional DL uses continuous target functions for regression and discrete target functions for classification. In contrast to traditional DL, DRL uses experience in an observable environment to decline a general error function and solve general problems. Self-learning control for energy-efficient transportation, adaptive emergency control, and voltage control are just a few of the control issues for which it is frequently employed. DRL is also used to solve optimization problems, such as learning the most effective bidding techniques in electricity markets, creating demand response plans for energy management, and determining the optimal cooperative schedules for wind and storage to lower uncertainty in renewable generation in smart grids. DRL has recently been introduced to power system data integrity defence, dynamic power allocation, and cyber-attack detection and recovery.

4.3. Classification of ML algorithms by subdomain

AI techniques vary in effectiveness depending on the specific subdomain of application within TESs. The following subsections provide a categorized overview:

4.3.1. Predictive modelling and performance forecasting

- **Algorithms Used:** ANNs, Support Vector Machines (SVM), RF, Extreme Gradient Boosting (XGBoost)
- **Application Context:** These algorithms are commonly used to predict system outputs such as outlet temperature, heat transfer rate, and collector efficiency based on input parameters like solar irradiance, ambient temperature, and flow rate.
- **Comparative Effectiveness:** XGBoost has shown superior performance in terms of prediction accuracy and computational speed due to its gradient boosting architecture. ANNs are flexible and effective for nonlinear systems but may require longer training times and careful tuning to avoid overfitting.

4.3.2. Control and optimization

- **Algorithms Used:** RL, ANFIS, Particle Swarm Optimization (PSO), GA
- **Application Context:** These techniques are employed for dynamic control of energy systems, such as regulating pump speeds, adjusting heat exchanger flows, or switching thermal loads to optimize efficiency in real-time.
- **Comparative Effectiveness:** RL, particularly Deep Q-Networks (DQNs), have been applied for self-learning controllers capable of adapting to environmental changes without explicit programming. ANFIS blends the learning capability of neural networks with fuzzy

logic reasoning. PSO and GA are suitable for multi-objective optimization, though they are often more computationally intensive.

4.3.3. Fault detection and predictive maintenance

- **Algorithms Used:** Long Short-Term Memory (LSTM) networks, Autoencoders, KNN, Decision Trees
- **Application Context:** Fault detection in components like heat exchangers, pumps, or photovoltaic thermal (PV/T) systems relies heavily on time-series data. LSTM models are particularly suitable due to their ability to capture temporal dependencies.
- **Comparative Effectiveness:** LSTM networks outperform traditional models in sequential data modelling but are more computationally demanding. Autoencoders are useful for anomaly detection and dimensionality reduction. Decision Trees and KNN offer fast classification but may lack precision in complex failure scenarios.

To synthesize the practical implications of applying AI techniques in TESs, Table 6 benchmarks commonly used algorithms in terms of their best application context, expected accuracy, and computational requirements. While fully normalized accuracy-to-computation ratios are dataset-dependent and thus omitted, the table offers a qualitative overview to guide selection based on performance vs. complexity trade-offs.

Emerging hybrid models that combine physical modelling with AI (i.e., physics-informed ML) offer promising improvements in interpretability and generalizability. Moreover, federated learning and edge AI are poised to enable distributed intelligence in thermal systems, facilitating faster and more secure decision-making at the device level.

4.4. Applications of AI in thermal energy systems

With its ability to temporarily store and reallocate thermal energy, thermal energy storage is a flexible technology with a significant impact on energy conservation and renewable energy sources. Its duties include thermal buffering, demand curve flattening, and energy output stabilization. Thermal energy storage helps bridge the supply-demand gap at night and during cloudy conditions by reducing the sporadic nature of solar radiation in concentrated solar power plants. Additionally, it can be used in wind farms to maximize the use of wind resources and in combined heat and power plants, which are usually used in conjunction with electricity generation and heat supply [187–189]. It is becoming more and more advantageous to combine thermal energy storage with AI. TES is essential for district energy networks, which use AI techniques

Table 6
A Concise Comparison of Key AI Algorithms Used in TESs.

Algorithm	Best Use Case	Typical Accuracy	Computational Load*	Notes
XGBoost	Performance prediction	High	Moderate	Fast, robust to overfitting
ANN	Forecasting, control	Mod.–High	High	Requires careful hyperparameter tuning
RL (DQN)	Real-time control	High	High	Learns optimal policies dynamically
LSTM	Time-series fault detection	Very High	Very High	Ideal for sequential data
PSO	Multi-objective optimization	Moderate	High	Effective for complex search problems
ANFIS	Adaptive control in uncertain conditions	High	Moderate	Balances accuracy and interpretability

* **Qualitative scale:** Low < Moderate < High < Very High (based on memory use, training time, and complexity).

to link different energy sources and consumers for environmental and economic advantages. AI, which simulates human cognitive processes like learning and problem-solving, is becoming increasingly well-liked because of its potential to increase efficacy and efficiency [190].

AI methods, like fuzzy logic and ANN, are inspired by natural phenomena, animal behaviour, and human brains. These methods are used in programming, optimization, operational control, and performance prediction. Through reasoning or modelling, AI generates robust solutions using approximate models. Although AI cannot produce flawless outcomes, it can handle ambiguity, imprecision, and partial truth, which makes it appropriate for challenging real-world issues. AI techniques save time by requiring fewer computational resources than physics-based model simulations like CFD. One important tool in using AI in thermal energy storage is fuzzy logic, which focuses on optimizing thermal energy storage design and operational control and predicting thermal energy storage performance. Solar water and air heaters, building HVAC systems, district heating and cooling systems, power plants, building envelopes, and PCM heat sinks for electronic cooling are just a few of the many applications of this technology [191]. Fig. 5 outlines AI's applications in thermal energy storage, focusing on performance prediction, design, and operational control optimization. Prior research articles have thoroughly examined the application of AI techniques in thermal energy storage.

While AI-driven optimization techniques offer remarkable improvements in predictive accuracy, control precision, and real-time adaptability of TESs, they also come with significant computational challenges that must not be overlooked. Advanced models such as deep neural networks, RL frameworks, and large-scale data-driven simulations demand substantial processing power, often relying on energy-intensive GPUs or high-performance computing infrastructure [190].

This computational overhead contributes to increased energy consumption, which can, paradoxically, undermine the sustainability objectives of the energy systems being optimized. Moreover, training these models requires access to extensive datasets and long convergence times, which may not be feasible for all applications, particularly in resource-constrained or decentralized environments. Therefore, careful trade-off analysis between performance gains and computational costs is essential, and future research should focus on developing lightweight AI models and efficient hardware integration strategies that minimize environmental and operational burdens while retaining optimization efficacy.

4.4.1. Predictive maintenance in AI-powered thermal energy system prediction

An approach known as “predictive maintenance” was created by IBM in the 1980s. It uses various data analysis techniques to find irregularities in equipment and procedures, spot possible flaws, and fix them before they become serious breakdowns [192]. The ability of AI to forecast TESs is transforming sustainability and energy management. Demand forecasting, system efficiency, and thermal energy production optimization have all shown promise with ML and DL approaches. AI and its learning techniques provide a quick and efficient means of problem-solving and scenario prediction using historical data. AI can greatly enhance thermal management by precisely forecasting circumstances and identifying the best solutions.

Several studies have adopted it in recent years to predict the performance of thermal energy storage. In order to predict the performance of a supercritical CO₂ Brayton cycle (s-CO₂BC) in combination with a wind farm-based molten salt energy storage system, For instance, [193] used AI. Utilizing the Grasshopper Optimisation Algorithm (GOA) and



Fig. 5. Diagram showing how AI techniques are being used in the thermal energy storage [191].

PSO, the Multilayer Perceptron ANN was trained. The complex interactions of molten salt energy storage, the wind farm, and the s-CO₂BC components are modelled by the AI models. Under various operational conditions, integrated system simulations were used to train the performance prediction model. Average absolute percentage deviations from the PSO approach ranged from 0.47 % to 0.51 % over a range of parameters, while those from the GOA approach ranged from 0.05 % to 0.27 %. The research boosts AI-based optimisation techniques and improves grid integration and stability. For the purpose of predicting the electrical and thermal efficiency of a water-based photovoltaic-thermal (PV/T) system, Safae et al. [194] developed ML models.

There are three ML algorithms that were selected because of their ability to interpret complex data and provide accurate predictions. These algorithms include the Gradient Boosting Regressor (GBR), the Multi-layer Perceptron (MLP), and the Light Gradient-Boosting Machine. Both the training and verification of the models were accomplished by utilizing data from 15,540 samples taken from a water-based PV/T system. The MLP method had the highest R² value of 0.0906, making it the most accurate predictor of electrical energy when compared to other GBR and LGB algorithms. This was proved by the LGB algorithm's R² score of 0.983, which demonstrated its effectiveness in estimating thermal energy. For the purpose of predicting the energy performance of a PV/T system that is based on an Ag/water nanofluid, various modifications were made to the most efficient MLP and LGB models. In spite of the fact that there was no actual data available, this indicated that the models were indeed valuable for practical applications. Furthermore, [195] created AI models to forecast thermal power plant emissions of nitrogen oxide. The AI model used in the study had more coal input features than boiler features, such as volatility, total moisture, fixed carbon, ash and sulphur content, and more. The study concentrated on the performance of AI models and the significance of the analysis of input features. The findings demonstrated that the number of features greatly impacts the model's performance. Eight distinct input features produced the best model performance. With an exceptional coefficient of determination of 0.97, the nitrogen oxide emissions prediction model for thermal power plants was accurate and feasible. High prediction precision was demonstrated by the calculated predictive values' close alignment with actual values.

Odufuwa et al. [196] demonstrated a predictive model that uses Ice Thermal Energy Storage (ITES) to forecast HVAC system performance. Indicators such as chiller power consumption, cooling thermal load, COP, and storage temperature were used in the model. Using nonlinear input variables to forecast ITES system configurations, the study employed an ANN model created using a shallow neural net fitting application. The model is trained, validated, and tested using performance metrics such as Mean Squared Errors (MSE) and Regression (R). The MSE was primarily less than 20 %, and the R values ranged from 0.94 to 0.99. The study emphasized the accuracy of ANN in predicting HVAC system parameters and the effect of parameter variation on ITES cooling systems. In another study, [197] analysed the NuScale VOYGR SMR module in RELAP5-3D, a Thermal Energy Storage system, and how it integrates and controls a Small Modular Reactor (SMR). The study utilised a two-tank thermal energy storage system and Model Predictive Control (MPC) to maximize the thermal energy storage system's performance in tandem with the SMR. Simulations, including accident scenarios, were used to evaluate the system's performance and response.

The California independent system operator database provided actual energy demand data, which the study scaled to account for the power generation of a single SMR. According to the results, a thermal energy storage system integrated with an SMR can improve performance compared to an SMR operating alone, but in some cases, the overall power mismatch may worsen. Arcos-Aviles et al. [198] demonstrated a model predictive controller-based energy management system for a remote electro-thermal microgrid in Ecuador's Amazon region. Mixed integer quadratic programming was used to formulate the optimization problem, and the FICO Xpress solver was used to solve it. Future

irradiance and load data were taken into account when running simulations. Analysis was done on how the prediction horizon affected the microgrid's performance. The findings demonstrated that operational microgrid cost, battery degradation, and temperature RMSE tracking decreased while optimizing PV resource utilization with increasing prediction horizon. Comparing the 24-h horizon to the 8-h horizon, the 24-h horizon decreased by 6.71 % and 15.68 %. The benefits of the MPC approach include a 31.57 % decrease in PV curtailment, a 5.86 % reduction in operating costs, better battery SOC planning, and an extension of the ESS lifetime.

Moreover, a new *in silico* thermal management approach for fuel cell systems utilizing GANs and RL was presented in the study by [199]. An RL agent can be trained to dynamically control system parameters using the realistic thermal management scenarios that GANs produce. The adversarial network can be trained using simulation results or historical data to produce a variety of scenarios illustrating the intricate connection between a fuel cell's thermodynamics and working point. In response to the generated working point, the RL agent adjusts cooling-influencing parameters like coolant flow rates, air stoichiometry, or reactant humidification. The method demonstrated notable gains in robustness, durability, and thermal efficiency. Also, [200]'s study investigated how ML could be used to maximize the efficiency of combined cycle power plants. It suggested an AI-coherent modelling strategy that concentrates on control and environmental factors. By adjusting optimal parameters and simulating multiple scenarios, the model enhanced efficiency using the Broyden Fletcher Goldfarb Shanno algorithm. Power generation increased from 452 MW to 462.1 MW, a 2.23 % increase. The study also emphasized the advantages of data-driven modelling, including its high accuracy, ability to improve with new data continuously, and ability to identify patterns that are challenging for human analysts. In the study by [201], the integration of RF, SVM, and ANN in three different kinds of solar prediction algorithms was examined. The findings demonstrate that, compared to SVM and RF algorithms, the ANN algorithm performed exceptionally well in terms of MAE (0.9558) and MFE (0.4466). This method can aid in planning and distribution in renewable power systems by tackling issues with intermittent and uncertain renewable sources.

Shaban et al. [202] study offered an AI-powered method for creating the best PV/T cooling systems. Preprocessing, prediction, and data collection were all part of the system. Water, air, and electricity efficiency was predicted using the Optimized Deep Neural Network (ODNN). The model's efficacy was increased using Sand Cat Swarm Optimization. According to the results, ODNN performs better regarding MAE, MSE, and R², indicating that it is a useful tool for designing PV/T cooling systems. The ability of the AI-driven method to maximize system operational efficiency was also shown. Finally, the study by [203] presented a technique for designing and optimizing thermal energy storage systems that combine ML and CFD. The approach trained ML models, such as K-nearest neighbour regression, linear regression, gradient boost regression, LightGBM, XGBoost, and neural networks, using heat discharging parametrizations from CFD. These models provide an effective substitute for conventional CFD simulations, allowing temperatures in compact thermal energy storage modules to be predicted under various circumstances.

Compared to full-scale CFD simulations, the ML-based method saves over 99 % of the computational time and costs, making it suitable for various thermal energy storage designs and geometries. Green roofs are becoming more and more significant in the design and retrofit of buildings for energy efficiency. Li et al. [204] introduced a new AI model framework that uses SHapley Additive Explanations and graph convolution to enhance their thermal performance. Using experimental data from green roofs outside the building and meteorological data from Shanghai, the model forecasts the surface temperatures of green roofs both inside and outside. Several metrics were employed to assess the model's predictive performance, demonstrating high accuracy. Based on the SHAP findings, the primary determinants of green roof thermal

performance are soil surface temperature and canopy temperature. Hajialigol et al. [205] investigated the application of organic Rankine flash cycles in conjunction with a Brayton cycle, emphasizing exergy, thermodynamic, and energy-economic analyses. Through the burning of methane from biomass gasification, the cycle produced thermal energy. The study also looked at several working fluids, such as toluene, ethylbenzene, m-, o-, and p-xylene. According to predictions made with an ANN, p-Xylene is the working fluid that most increases energy efficiency. By 10 %, the enhanced cycle reduced energy destruction. Nevertheless, double flash evaporators increased energy efficiency by 3 % while raising the cost of power generation units by more than 10 %.

Kijo-Kleczkowska et al. [206] created a model based on fuzzy logic to forecast the highest amounts of heat generated by materials under test at particular temperatures. They postulated that maximum heat values and pollutant emissions are reflected in the calorific value and composition of the materials and that the model predicts maximum heat values during the thermal processes under analysis. The study by [207] considered heat source and thermal slip effects while analysing heat transfer in Williamson fluid flow using AI. Equations of momentum and energy were used to formulate the flow. The neural network model with the best validation performance, 5.9676×10^{-8} , was trained using the Levenberg-Marquardt algorithm. The study also found that the Nusselt number exhibited decreasing trends, whereas the heat generation and thermal slip parameters were higher in non-magnetic flow fields.

Despite the promising accuracy of AI models in predicting TES performance, a critical limitation lies in their frequent reliance on steady-state or average-condition datasets. Such datasets often fail to capture the dynamics of extreme operational scenarios (such as abrupt load surges, component failures, or climatic extremes) which are increasingly common in real-world applications. As a result, models trained solely on nominal data may yield inaccurate or unstable predictions when exposed to atypical but realistic conditions. To enhance model robustness and adaptability, future development should integrate transient and extreme-case datasets, possibly through data augmentation, synthetic stress-test simulations, or real-time adaptive learning algorithms. Furthermore, combining AI with physics-informed modelling can provide better generalization and improve reliability, especially under non-linear and rapidly changing boundary conditions. This approach is essential for deploying AI-based TESs in complex and uncertain operational environments [204].

For AI models to be practically valuable and scientifically credible in thermal energy optimization, reproducibility and transparency are essential. However, many current studies [206,207] lack publicly accessible code repositories and omit key algorithmic details, such as the number of hidden layers, learning rates, or optimization functions used in neural networks. These omissions hinder experimental validation and slow progress in comparative benchmarking. To promote reproducibility and collaborative advancement, future research should document algorithm configurations comprehensively and make implementation code and datasets openly available through repositories such as GitHub or institutional archives. Encouraging open-access frameworks will enhance the scientific integrity, replicability, and scalability of AI-assisted TESs.

4.4.2. Energy consumption forecasting

Forecasting energy consumption is a crucial process in energy management that aids in the informed decision-making of utilities, businesses, and governments by estimating future usage based on past data and influencing factors. In energy consumption forecasting, AI-based models such as ANN, Support Vector Regression (SVR), and RF are frequently employed; ANN and SVR are the most widely used and well-liked models for resolving various forecasting issues. An ANN is a framework for various machine-learning algorithms that cooperate to process complex data inputs. ANFIS, Feedforward Neural Networks (FFNN), BackPropagation Neural Network (BPNN), wavelet neural networks (WNN), ESN, DL models, and others are ANN-based models

used in energy consumption forecasting. By analysing and processing complex data inputs, these models improve the efficacy and efficiency of forecasting energy consumption [208–210].

Two kinds of ANN are used in energy consumption forecasting: FFNN and BPNN. FFNN moves forward information from input nodes to output nodes via hidden nodes. Using a back-propagated algorithm, BPNN spreads the error difference between the calculated and desired outputs toward hidden layers. These models have been called conventional ANN models in studies on forecasting energy consumption. Szoplik [211] and Akdemir et al. [212] used a three-layer FFNN to estimate Poland's and Turkey's energy demands. Over the past several years, a number of studies have been carried out for the purpose of predicting the future energy consumption of a variety of facilities or sectors, including countries. These studies have made use of a wide range of AI systems.

On the basis of the difficulties of high systematic errors and poor generalization that are present in combination Forecasting Models (FMs) that are currently in use, [213] presented a dynamic combination residual FM as a solution to these issues. This model is depicted in Fig. 6. For the purpose of predicting the residual errors, it was agreed that support SVR would be used as the basic FM, and the dynamic combination FM that was developed on the basis of weights would be deployed. A subsequent modification was made to the value of the SVR predicted after that. With regard to the ideal number of combination terms as well as the selection of the single FM in the combination model, this work offers a mathematical basis for both of these decisions. The proposed FM has the potential to reduce the Mean Absolute Error (MAE) of the basic model from 1918.59 kW to 349.37 kW, the Mean Absolute Percentage Error (MAPE) from 15.80 % to 2.96 %, and the Root Mean Square Error (RMSE) from 2278.74 to 471.44 according to the findings of a case study that was carried out in Xi'an. The study was conducted to identify the limitations of the FM.

To forecast energy statistics in the Chinese province of Shandong, Li and Li [214] made use of the Generalised Modelling (GM) approach. This was done to make predictions. An enhancement in the predictability of the original prediction was achieved as a consequence of the utilization of the ARIMA model for forecasting residuals. The combined GM-ARIMA model was shown to be more accurate than the separate models; this was discovered by a comparison of the average relative errors. The differences between the two models served to illustrate this point.

KNN is a popular prediction algorithm because it is easy to use and straightforward when locating similar instances in large-dimensional and multivariate feature spaces. However, since it only pinpoints historical, it is not a causal approach to forecasting. To enhance forecasting, temporal data such as the day of the week or the week of the year should be added to KNN [215]. Several studies have used this algorithm to predict the performance of energy systems. For instance, through precise analysis of energy requirements, the paper by [216] sought to manage energy production, storage, and management in the residential sector in an efficient manner. To predict daily energy consumption based on classification, the KNN classifier was used. Prediction, validation, processing, and data collection were all steps in the process. Experiments using historical data from 520 apartments in Seoul showed that the best accuracy was achieved at a training and testing ratio of 60–40 %. The model's effectiveness was verified through the use of 5-Fold and 10-Fold cross-validation. Similarly, a surrogate modelling technique was presented by [217] to forecast hourly cooling/heating loads for office, hotel, and retail buildings in high resolution throughout the year.

In EnergyPlus, it generated a load database with seven building features and hourly loads using 16,384 surrogate models. KNN was selected as the data-driven load prediction algorithm. It was discovered that the best metric for effective hour rates was the Manhattan distance. By forecasting thermal loads in Shanghai, China, the method was validated. MAPEs for cooling/heating and annual thermal loads were 5.26 % and 6.88 %, respectively. Ramos et al. [218] investigated how ANN and KNN might be used to forecast building energy usage trends.

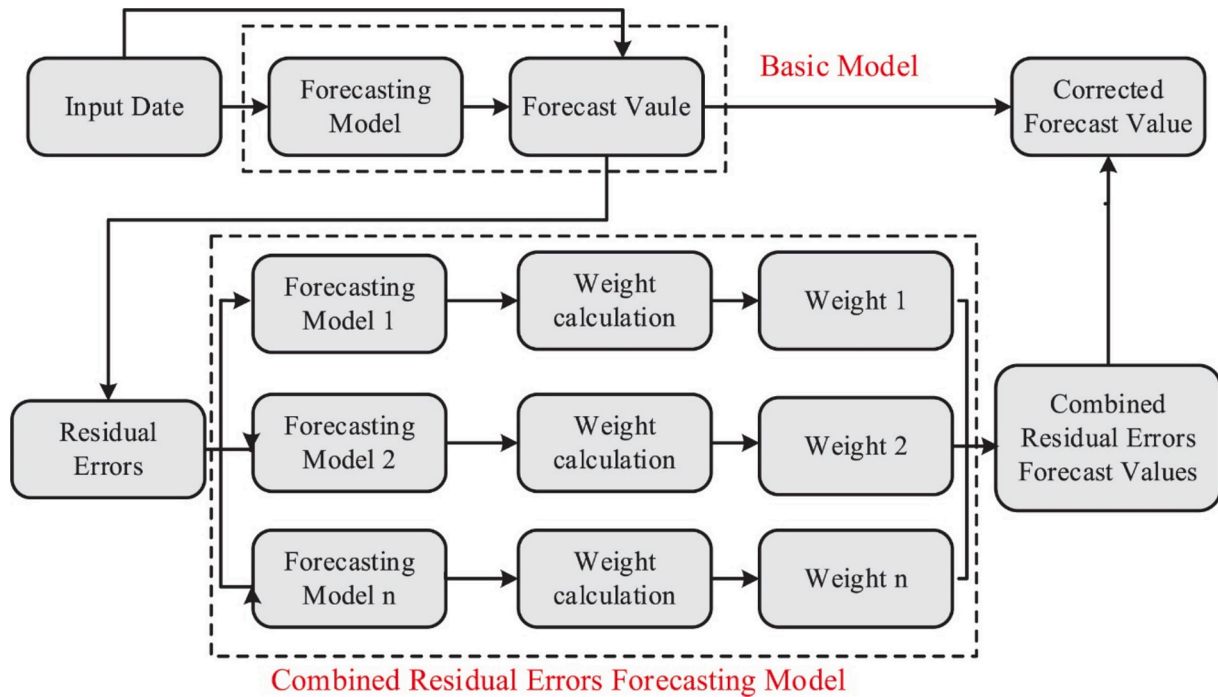


Fig. 6. FM framework with combined residual error correction [213].

Considering the forecasting technique to enhance predictions, the study employed five-minute forecasts. To increase forecast accuracy, the Multiarmed Bandit in RL was utilised to determine the optimal algorithm for every five minutes. The study evaluates greedy algorithms and upper confidence bounds in three different scenarios. The findings demonstrated that although ANN has a lower confidence bound, KNN is the most reliable forecasting algorithm. KNN was more potent than greedy, and the average rewards demonstrated sound decision-making for morning, afternoon, and night to increase the accuracy of short-term electric load forecasting for individual users, [219] proposed a short-term power load FM based on K-means and FCM-BP. Using K-means, users were grouped into two clusters based on their electricity consumption characteristics.

Users with strong correlation at adjacent moments were filtered out of local similar data using fuzzy C-Mean clustering. Users with weak correlation were selected using locally similar daily data. The BPNN uses feather vectors as input data to predict short-term load. According to experimental results, the suggested clustering approach more accurately forecasts loads than non-clustering load forecasting because it aligns with users' electricity consumption patterns. To minimize energy consumption by continuously monitoring building thermal comfort values, [220] demonstrated a thermal comfort monitoring system that replicated commercial building occupants in hot-humid climate nations. The system employed six thermal comfort factors and a KNN model. With an accuracy rate of 72.97 % in Malaysian university lecture halls and 35.13 % in non-hot humid climates, the model outperformed PMV in both hot and non-hot humid climates. Furthermore, SVRs have been employed in demographic studies, like those conducted by Wei-Chiang Hong [221] and Cao & Wu [222], to examine the growth in electricity consumption. The study conducted by [221] uses historical data from Taiwan's regional electric load values to compare SVR with other techniques, such as neural networks and regression models. In their work, Cao and Wu [222] use monthly data on China's electricity consumption, employing a hybrid approach to improve the analysis of electricity consumption and prevent over-selection of parameters.

Zhong et al. [223] presented a SVR technique based on vector fields that achieve high robustness, accuracy, and generalization ability. When the technique was tested on a sizable office building in China, it

outperformed other widely used techniques. Magoulès et al. [224] investigated a different algorithm for predicting building energy consumption using support vector machines. A year-long dataset was used for the experiments, and weather and electricity use were used as independent variables and parameters, respectively. After three months of testing and validation, the model showed good performance. Similarly, the SVR-supervised ML method was utilised in the research conducted by [225] to evaluate the ways in which eight primary input elements influenced the HVAC loads in residential structures.

Out of the six meta-heuristic optimisation methods that were investigated, the hybrid model SVR-AEO displayed the greatest performance in terms of modelling heating and cooling loads. Its R^2 values were 0.9975 and 0.99955, respectively. For the purpose of predicting the amount of useful hourly energy that can be extracted from a solar thermal collector system, a study was conducted by [226] that compared decision trees (DT), tree-based ensemble ML models (RF and ET), and SVR. A comparison was made between them based on the correctness of the models, the amount of computing cost, and the potential for generalization. Regarding the prediction of useful solar thermal energy, RF and ET were equally applicable and possessed similar predictive power. DT required less training time but was less accurate, making it the most computationally efficient approach. Training times for SVR were three times longer than those for ET, suggesting that DT is more accurate than RF and ET. The relevance of predictive analytics in the management of decentralized energy systems is emphasized throughout the paper. In the research conducted by [227], a ML model was provided as a means of predicting the amount of energy that is utilised for heating residential buildings. SVR was utilised as the basis, and hybridisation strategies were implemented in order to integrate six different meta-heuristic algorithms.

For the purpose of evaluating the accuracy and performance of the technique in comparison to hybrid models, a case study was utilised. Based on the findings, it was proved that the proposed method accurately predicts the amount of heating energy that is consumed by residential structures. The hybrid SVR-Battle Royale Optimisation model performed the best out of all the hybrid models that were investigated. Its R^2 values for the train dataset were 0.999386, and its R^2 values for the test dataset were 0.998898. Another tool is the Gaussian Process

Regression (GPR). Compared to other techniques, GPR is a more generalised ML model that performs exceptionally well in Bayesian inference about functions, especially in situations where there is insufficient training data [228].

García-Nieto et al. [229] introduced the GPR approach, a nonparametric Bayesian technique to forecast energy performance in residential buildings in oceanic climates. With a coefficient of determination of 0.9687, the model successfully predicts the total power output and determines the significance of each predictive variable. The effectiveness of this novel strategy is demonstrated by the agreement between experimentally observed data and predicted data from the suggested hybrid DE/GPR-relied model. Another study by [230] examined hourly HVAC system energy consumption and meteorological data using a government office building in Ningbo as a case study. It predicted the short-term hourly energy consumption of building HVAC systems using a 12×12 grid search and GPR. The optimal model's CV-RMSE was 0.1035 and 0.1278, and its R^2 was 0.9917 and 0.9863. Compared to ASHRAE standards, the NMBE was more accurate at 0.0575. Also, a paper by [231] examined the differences between using a grey-box model and Gaussian processes for thermal building modelling.

A single-zone office building with hydronic heating served as the model. The Gaussian processes outperformed the grey-box model in prediction errors during occupied times after three weeks of training by 27 %. However, their large generalization error caused them to perform worse during unoccupied times consistently. The Gaussian processes were trained adaptively to lessen the effect of weather variations. Every 24 h, the models were retrained, which decreased prediction error by more than 21 % and 10 %, respectively. However, the model currently covers only applications where prediction during occupied times is more relevant. By examining the impact of temperature on powertrain efficiency and auxiliary power, [232] also forecasted energy consumption at the bus trip level. Taking driving style, spatiotemporal features, and speed state transition into account, a Markov-based speed profile generation technique was created. The predictions' accuracy was increased using a Markov-based Gaussian Processing Regression Model (M-GPR). With an average error of 8.69 %, which decreased by 0.62 % to 5.41 % compared to other models, the results demonstrated that M-GPR performed better in prediction.

RFs is a ML model that is commonly used for producing predictions because to its high success rate in categorising a large data set. However, it requires temporal variables in order to function well [233]. When it came to predicting the amount of energy that was consumed by HVAC systems in a hotel located in Madrid, Spain, [234] contrasted RF and ANN. An ANN fared marginally better, with root-mean-square errors of 4.97 and 6.10, respectively. Ensemble-based methods, on the other hand, delivered superior performance when dealing with complex data in buildings, despite the fact that RF had internal cross-validation and a limited number of tuning parameters. The ability of both models to make accurate predictions was comparable, and they were approximately equally applicable to applications involving building energy, estimating the amount of energy that was utilised to heat dwellings was accomplished by Xu et al. [235] through the utilization of the RF method and meta-heuristic optimisation approaches. All of the following factors were taken into consideration: occupancy, ambient temperature, and architectural elements. RMSE, MAE, Relative Absolute Error (RAE), Theil Inequality Coefficient (TIC), and coefficient of determination were the five statistical indices that were utilised in order to evaluate the effectiveness of the model.

The findings demonstrated the accuracy of the model's ability to forecast energy consumption, and the performance of RF was significantly enhanced by the application of the Arithmetic Optimisation Algorithm. Following a total of 101 rounds, the Ant Lion Optimiser achieved a TIC value of 0.014986, demonstrating remarkable convergence throughout the process. The suggested combination model did a much better job of predicting how much energy homes would use for heating than traditional methods. In the same way, [236] used Quantile

Random Forest (QRF) to guess how much power a chiller plant in a high-rise office building would use. A streamlined data-driven method was used to figure out the lower and upper limits of the power usage prediction intervals (PIs). The results showed that all chiller models had PI coverage probabilities above 80 % and 90 %, along with the confidence levels that went with them. The range of real demand charges was 3.3 times wider than the upper and lower bounds. This difference grew by 1.7 times when the confidence level went from 80 % to 90 %. In addition, the study by [237] showed a prediction model based on RF that could guess how much energy different buildings would use in the next few days. Five sets of data from over a year were used to test the model. The results showed that the MAE number was between 0.430 and 0.501 kWh, which means that the predictions were pretty accurate.

4.5. Challenges associated with AI predictions in energy systems and future directions

Numerous difficulties arise when using AI to forecast energy systems, including the two main limitations of AI techniques in TESSs: the lack of data available for establishing prediction models and the inability to reveal unknown physical knowledge and mechanisms directly [191]. The performance of AI systems and accurate predictions depend on high-quality data because noisy, out-of-date, or incomplete data can produce unreliable models and make training more difficult [238]. Also, AI models in energy systems struggle with scalability due to the vast amount of data produced, making successful implementation challenging across extensive operations. Finding complete, high-quality datasets is another major obstacle to AI energy system prediction. Large amounts of well-structured data must be available for these models to be accurate. On the other hand, inaccurate data can result from incomplete, noisy, or fragmented data. Using efficient data collection and preprocessing techniques reduces this problem and enhances model performance [239].

Furthermore, compatibility problems and the need for real-time processing make integrating AI solutions into energy management infrastructures difficult. Standardized and modular architectures are required for these systems to operate smoothly and perform better. Operators find it challenging to comprehend complex AI models, such as DL models, because they struggle with interpretability despite frequently delivering high accuracy. This lack of transparency may make Critical energy systems difficult to adopt [240,241]. Research recommends hybrid models combining domain expertise and AI to enhance interpretability and support better decision-making. A multi-disciplinary strategy integrating AI technology developments and energy system knowledge is required to address obstacles in energy systems. The future of energy management will be shaped by the cooperation of academia, business, and policymakers to fully realize AI's potential for efficient and sustainable energy systems. AI integration in TESSs is a quickly developing field with many potential research avenues, including important areas for further investigation. Some of these areas include:

- Future research can examine how AI-driven methods can optimize TESSs in real-time, emphasizing algorithms that adjust to shifting supply and demand.
- Future research can also employ RL and sophisticated AI techniques to create data-driven control strategies to optimize the operations of TESSs.
- Future research can examine how AI methods can optimize district heating networks to increase effectiveness, cut expenses, and minimize environmental effects.

The current global energy consumption scenario can be characterized by four major aspects: rapid growth in overall energy consumption, high dependence on non-renewable resources, unequal distribution of fossil fuels, and considerable geopolitical and economic barriers

[242,243]. In this context, optimizing energy performance has become an important social, political, and economic factor in every country. A prominent, innovative and disruptive factor that can enhance the performance and efficiency of TESs is the use of AI and ML, especially when associated with control systems. Renewables, especially solar and wind energy, are complementary and essential solutions to bridge traditional and non-traditional limitations.

Although several insist that current and future sources of renewable energies are capable of meeting global energy demand, a few challenges need to be addressed in the thermal energy sector, such as a lack of production time or available space [244]. In this context, some challenges with thermal energy need to be addressed, such as energy storage, microgrids, or hybrid configurations with energy backup [245,246]. The promising potential of nanotechnology is an important aspect in meeting this challenge, as it is in the traditional arena of improving heat transfer, deposition, and efficiency of thermal energy [247]. However, nanomaterials are also promising solutions for energy storage and other specific requirements for many advanced configurations of renewable and biothermal systems in which thermal effects must be addressed [248,249].

5. Synergy of nanotechnology and AI in optimizing thermal energy systems

The synergy between nanotechnology and AI is transforming the optimization of TESs [250]. When combining nanotechnology with AI's intelligent and data-processing aspects, energy systems are becoming more flexible, innovative, and self-regulating [251]. This integration allows real-time control and optimization, which has greatly improved how energy is produced, stored, and regulated [252,253].

5.1. Integration of nanotechnology and AI

Nanotechnology and AI are the two most powerful technologies changing the approach to managing TESs and energy generation, storage, and regulation. These systems are gradually becoming self-optimizing by integrating the physical progress of nanotechnology with the analytical and prognostic features of AI. This integration of two systems allows for real-time control and maximization of the benefits of nanomaterials enhanced thermal conductivity while receiving guidance from AI [254].

This subsection elucidates the intricate mechanisms through which nanotechnology (NT) and AI synergize to optimize TESs. AI algorithms, including ML and advanced optimization techniques, significantly accelerate the discovery, design, and synthesis of novel nanomaterials possessing tailored thermal properties. Conversely, the distinctive capabilities of nanomaterials (such as their expansive surface area, enhanced thermal conductivity, and tuneable properties) provide unprecedented avenues for developing more efficient and intelligent AI-driven thermal management solutions [253]. For instance, AI-guided synthesis facilitates the creation of advanced phase-change materials, ML models accurately predict nanofluid performance, and AI optimizes the designs of nanostructured heat exchangers, collectively showcasing these powerful synergistic effects [254].

A critical component of this section is the economic quantification of these NT-AI synergies. This analysis details the financial benefits and implications stemming from the integration of these advanced technologies. It encompasses potential cost reductions in both material synthesis and system operation, significant energy efficiency gains translating into substantial savings, and robust projections for return on investment (ROI) across various NT-AI applications within thermal systems. Furthermore, the discussion highlights market growth projections for these converged technologies, identifying new economic opportunities and assessing the feasibility of their large-scale implementation. Supporting this analysis are relevant economic data, detailed industry case studies, and future projections, which collectively

underscore the compelling economic rationale behind leveraging NT-AI synergy for sustainable and highly efficient thermal energy management [255].

5.1.1. Conceptual framework for integration

The framework for integrating nanotechnology and AI relies on two key components: nanomaterials and artificial intelligent control systems. Nanofluids, NePCMs, and nano-coatings are used to improve the thermal performance of the systems by increasing heat transfer and storage rates [255]. For instance, the nanofluids containing copper or graphene nanoparticles exhibit significantly enhanced thermal conductivity than conventional fluids for application in heat exchangers and refrigeration systems. AI supports these materials by providing real-time tracking, data processing, and adaptive decision-making. Such ML applications include systems' ability to predict thermal loads, automatically optimize the operating parameters, and achieve the best energy efficiency. In solar thermal systems, AI can control nanofluids' flow rate and maintain the heat absorption and transfer rate despite the changes in the external environment [13].

5.1.2. Key technologies and techniques

Several technologies and techniques play a crucial role in successfully integrating nanotechnology and AI within TESs:

- **Nanofluids:** Base fluids can be water or ethylene glycol, and nanoparticles added are aluminium oxide (Al_2O_3) or silicon carbide (SiC). These nanoparticles enhance the thermal conduct and storage capacity to enable heat transfer at a faster rate. Based on the current system requirements, AI-based models control the flow and temperature of nanofluids [256].
- **Nano-Coatings:** Nanoparticle coatings in thin layers on heat exchanger surfaces minimize fouling and improve heat transfer. These coatings also have anti-corrosive and self-cleaning features, requiring little or no maintenance. Nano-sensors are used to observe signs of fouling in real-time and thus adjust cleaning cycles to guarantee constant high performance [257].
- **AI Algorithms:** Multilevel AI, such as neural networks, RL, and GA, enhance system performance by analysing Nano-sensors' massive data. These algorithms forecast system performance, estimate the energy requirement, and control parameters proactively. For instance, RL allows systems to learn from previous performance and optimize energy use in real time [258].

5.1.3. Practical impact

Novelties in nanotechnology and AI provide a potential for improving the efficiency of the otherwise conventionally run TESs. These working fluids in the systems are enhanced by incorporating nanotechnology particles in heat transfer processes, and the systems are then controlled in real-time by AI. The outcome is quicker heat exchange, higher combustion or electrolysis rates, and faster system starts without subjecting the system to dangerous conditions. While most thermal energy processes are analysed under steady-state conditions, which imply continuous and constant operation, non-ideal transient conditions frequently cause performance problems; with the help of nanotechnology AI provides solutions to these challenges by enabling systems to attain a steady state faster and more effectively. When enhanced with nanotechnology and phase change physics, the fundamental thermodynamics equations enhance understanding of the system behaviour and significantly improve system performance [259].

5.2. Case studies of combined approaches

The case studies in this section demonstrate the application of AI tools to enhance surface modification processes at the atomic level and enhance the performance of several components in TESs. They are heated surfaces, silicon carbide heat transport fibres, contaminated and

evaporating porous surfaces, and crystal growth systems. Various research units conduct studies in a given institution to strengthen the technologies in these important fields.

Many recent approaches utilizing AI are explored as follows:

- *Optimization of Nanoscale Structures:* This strategy uses AI tools to optimize nanoscale structures as a design strategy. In earlier studies, genetic or evolutionary algorithms have been integrated with detailed performance simulations to accomplish these optimizations. The performance enhancement is qualitatively determined by parameters like thermal conductivity and heat transfer coefficient; hence, the enhancement is quantifiable [260].
- *Surface Modification Choices:* In this approach, AI helps make decisions about surfaces that affect several micro/nanoscale phenomena across the material. GA are used to control thermal variation and surface contaminant behaviour at the same time. The effectiveness of these changes can be assessed by experimental verification, which involves thermodynamic efficiency tests and monitoring of the parameters during operation [261].

5.2.1. Nanofluid-based heat exchangers with AI control

The first case study is based on a heat exchanger system employing a nanofluid in an industrial cooling process. This system exploits a copper nanofluid to improve the thermal conductivity of the coolant used in the system. An AI system constantly observes key process parameters, including temperature and flow rate, through nano-sensors. The flow rates in the system can then be controlled by the AI using ML algorithms that allow the system to adapt the flow rates in real-time to changes in heat load. This flexibility has translated into a 20 % increase in energy efficiency and has been done without the need for operators to get involved. Energy consumption indicators were used to measure the efficiency improvements, showing how the system improved performance under different operating conditions [262].

5.2.2. AI enhanced solar thermal energy with nanomaterials

The second case study focuses on using AI optimization in a solar thermal system using nano-coated solar absorbers. These absorbers, TiO₂ nanoparticles coated, enhance the system's absorption capabilities and convert solar energy into heat. The AI system is used to track environmental factors such as the intensity of the sun and the temperature of the surrounding air, and with the help of a programmed ML algorithm, the system determines the best settings for the operation of the installation and the angle of the solar panels. This system orients the solar panels to the proper angle and controls the circulation of nano-enhanced heat transfer fluids; the energy efficiency is 25 % greater than traditional systems. These enhancements were confirmed by field tests where energy, produced, and conversion efficiencies were quantified and compared to standard solar thermal systems [263]. In summary, this work has shown that applying nanotechnology and AI in TESs can significantly improve the thermal performance and the systems' general performance. With the help of micro/nanoscale techniques and engineering knowledge, the presented approach can be implemented in several sectors, such as renewable energy, the aerospace industry, manufacturing, and others. The derivations lead to enhanced energy practices that comply with international energy efficiency standards and facilitate advancement in energy technologies.

5.3. Benefits of AI-nanotechnology integration

The integration of AI with nanotechnology, which is another technology that is quickly expanding, is one of the most promising areas of development. Nanotechnology has various benefits that can increase the performance, efficiency, and efficiency of TESs. The author examines the most significant advantages that this relationship, which is advantageous to both parties, offers in this section [264].

5.3.1. Increased efficiency and performance

1. *Improved Thermal Conductivity:* Nanofluids and nano-enhanced phase transition materials are examples of the types of materials that have been developed via the use of nanotechnology. In the processes of thermal energy conversion, the utilization of these materials leads to enhanced heat transfer coefficients, which in turn leads to increased efficiencies.
2. *Real-Time Optimization:* Another advantage of AI is that it enables constant tracking and control of the parameters of a system in real time. AI, using data obtained from the sensors placed in the system, can control the conditions under which the system operates, hence the efficient use of energy resources and, therefore, the reduction of energy use and cost.
3. *Predictive Maintenance:* Intelligent technologies enable predictive maintenance methods, which utilize analytical information to anticipate failures in TESs. If problems could be foreseen, maintenance could be planned to prevent scheduled breakdowns, thus reducing the frequency of maintenance and its expenses.

5.3.2. Cost savings

1. *Reduced Operational Costs:* Integrating AI systems into operations improves efficiency and optimality, thus lowering operational costs. Hence, reducing energy use and increasing the system's efficiency will greatly reduce costs in the long run.
2. *Lower Maintenance Costs:* Applying different elements of AI to predictive maintenance plans can help minimize how often a system suffers from a failure and the extent of the failure. As a result, the maintenance cost is reduced, so it is possible to optimize the distribution of funds.

5.3.3. Sustainability and environmental benefits

1. *Reduced Energy Consumption:* AI-nanotechnology integration leads to optimized TESs' performance, reducing energy consumption. This reduction is instrumental in reducing the emission of greenhouse gases, hence leading to a sustainable energy environment.
2. *Utilization of Renewable Energy Sources:* The integration improves the effectiveness of solar TESs, leading to using renewable energy sources. The effectiveness of these systems can be enhanced to reduce dependency on fossil fuels, hence enriching the use of clean energy.

5.3.4. Improved system flexibility

1. *Dynamic Response to Environmental Changes:* AI helps TESs respond to changes in environmental factors such as temperature and humidity. It makes the system's performance very reliable because it can work at its best despite the prevailing conditions.
2. *Customization and Flexibility:* The integration enables one to create solutions that can be customized to suit a particular use and setting. AI algorithms can capture unique operation parameters and thus develop systems that efficiently meet certain energy requirements.

5.3.5. Research and Development

1. *Accelerated Innovation:* AI and nanotechnology complement each other to enhance the fast development of research and innovation. Through simulation and modelling by AI, time is saved in experimentation and testing, hence the quick development of thermal energy technologies.
2. *Interdisciplinary Collaboration:* The integration fosters working in teams and model experts from materials science, thermal engineering, and data analytics. It fosters the exchange of knowledge and can result in solutions that are unique and take advantage of the peculiarities of the disciplines involved.

5.4. Challenges and barriers to integration

Integrating nanotechnology and AI in TESs presents a promising avenue for enhancing efficiency and performance. However, several challenges and barriers must be addressed to realize the full potential of these technologies. This section outlines the primary obstacles hindering their successful integration.

5.4.1. Technical challenges

1. **The complexity of Nanomaterials:** This behaviour is because the properties of nanomaterials are quite different from the material's properties at the macro scale. This complexity requires the development of enhanced characterization methods and using models to estimate their behaviour in TESs. A lack of standard assessment procedures can lead to doubt concerning using such materials.
2. **Data Quality and Quantity:** AI systems require large amounts of high-quality data to train and make decisions for them to be effective. However, information obtained from nanotechnology applications may be scarce, incoherent, or of low credibility. This scarcity can limit the creation of accurate AI models crucial for enhancing TESs.
3. **Integration of Multi-Scale Models:** Coupling nanoscale phenomena with macroscale thermal systems requires consistent multi-scale modelling strategies. Constructing models of how nanomaterials interact with larger systems is challenging and may require computational resources.

5.4.2. Economic barriers

1. **High Development Costs:** The R&D period may be expensive when integrating nanotechnology and AI. Lack of capital may limit the feasibility of research institutions and companies engaging in innovative projects, especially for small institutions with inadequate capital to invest in the projects.
2. **Market Readiness and Commercialization:** It is not always easy to scale up innovations made in the laboratory to true commercial products. This situation means that industry stakeholders may avoid adopting new technologies with no prior market standard or a track record of success. This, coupled with the long testing, validation, and certification period, will only add to the slowdown of the commercialization of integrated technologies.

5.4.3. Regulatory and safety concerns

1. **Regulatory Frameworks:** The current and future legal frameworks regarding utilizing nanotechnology and AI raise the question of developers' risk. This lack of specificity may impede the implementation of such technologies into recognized structures. Such regulatory agencies need to set the right benchmarks on how nanomaterials and AI in TESs can be used safely to avoid harming society.
2. **Health and Environmental Impacts:** Possible hazards that nanomaterials pose to the environment and the people in the community must be examined. Some of the issues, such as the safety of nanoparticles and the effects of nanoparticles on human health, can be considered major challenges for nanoparticle technology distribution.

5.4.4. Interdisciplinary cooperation

1. **Siloed Research and Expertise:** There is a need to collaborate with material science and thermal engineering experts, AI, and other related experts to integrate nanotechnology and AI effectively. However, such efforts are typically isolated, limiting interaction and cross-disciplinary information sharing.
2. **Training and Skill Development:** This is due to the fast-growing nature of the nanotechnology and AI industries, which require constant

continuing professional development for its workers. The skills researchers and practitioners should have to work with these technologies are essential, so a great investment in educational programs is needed.

6. Recent advances and innovations

Renewable energy sources such as solar and wind energy are inexhaustible. Still, their availability depends on the presence of wind and daylight hours to harness solar energy, and therefore, they are subject to periodic fluctuations. These fluctuations pose challenges that prompt researchers and stakeholders to search for energy storage and management solutions. One of the methods used recently to develop thermal storage, charging, and discharging technologies is the integration of PCMs with nanoparticles such as Al_2O_3 and CuO , in addition to introducing AI technologies to achieve the best energy management and utilization. These advanced technologies are not limited to renewable energy applications only but also to all applications related to thermal energy, such as buildings and heat exchangers, to achieve sustainability and provide energy.

6.1. Cutting-edge research in nanotechnology for thermal energy

The technology of integrating nanoparticles into PCMs (Called NePCM) has recently taken the largest place in the interests of researchers compared to other technologies, such as the technology of the nanofluids used to maximize thermal performance [265]. Arıcı et al. [266] presented a numerical investigation to improve the PCM melting process in a square cell using variable size and position fins and CuO nanoparticles. The charging process is carried out by applying a heat flux to the two walls while the other remains thermally insulated. Their results indicate that CuO nanoparticles in PCM significantly enhance the charging rate even with small fins. In the thermal energy storage unit as

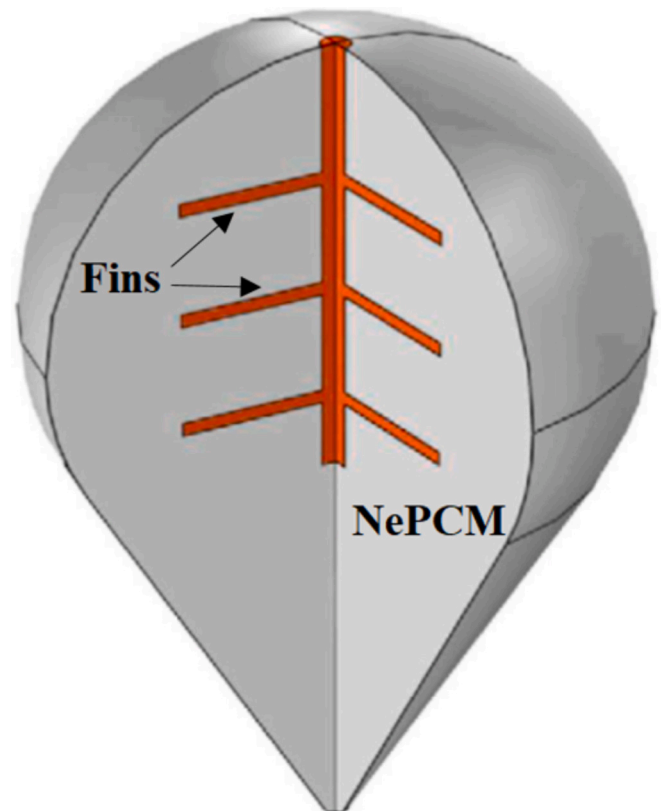


Fig. 7. Pear-shaped and finned container [267].

pear-shaped and finned, as shown in Fig. 7.

Murad et al. [267] conducted a numerical analysis on the phase change process of NePCM containing Al_2O_3 nanoparticle. The equations governing heat transfer were numerically solved using the finite element method enthalpy-porosity method. Their findings pointed out that the increase in the concentration of Al_2O_3 nanoparticle nanoparticles improves the melting rate because adding 6 % of Al_2O_3 to the PCM shortens the melting time by 12.5 %. They also suggested using thinner and longer fins to increase the melting rate since heat diffusion in the far regions of the thermal storage unit was also high. Basem [268], in his recent study, conducted a numerical study to demonstrate the charging of a semi-cylindrical cell-filled paraffin RT58 as PCM. The vertical straight-walled cell has been exposed to a constant heat flux by circulating hot pure water and hot nanofluid (Al_2O_3 -water), and the curved wall is fully insulated. Enthalpy and porosity were employed quantitatively using the ANSYS/FLUENT 22 program to analyse the study. The results also indicated that the complete melting time of paraffin was 3 %, 4 %, and 5 % less when using 10 %, 20 %, and 30 % nanofluid concentrations than water. In thermal applications such as heat exchangers.

Motevali et al. [269] conducted a numerical study to investigate the melting process of PCM in a double-tube heat exchanger. RT82 was used as the PCM, and SWCNT as the nanoparticles. The concentrations of NePCM were 1, 2, and 3 %. The findings provided by the authors showed that integrating NePCM in the heat exchanger improves thermal performance, decreasing the melting process by 39 %. Ghalambaz et al. [270] have suggested CuO and GO as nanoparticles for improving NePCM integrated with a layer of copper foam to optimize the thermal performance of the shell-and-tube Latent Heat Thermal Energy Storage (LHTES). To improve thermal conduction around the tube and reduce interference with the suppression of convection, the copper foam was arranged in an irregular geometrical pattern around the tube, as illustrated in Fig. 8, while the remaining part of the shell was packed with NePCM.

Heat transfer was modelled by solving the partial differential equations of conservation of both momentum and energy. In their respective experiments, they found that the nanoparticles added minimized the melting time by 12 % when compared to PCM alone. Furthermore, it was also observed that CuO nanoparticles were marginally better than GO nanoparticles in heat transfer enhancement. Zahid et al. [271] conducted an experimental study of heat sinks for cooling electronic devices, comparing three types: straight, round, pin-finned heat sinks and copper foam heat sinks. They were evaluated against NePCM containing RT54HC/ Al_2O_3 , with the percentage of Al_2O_3 ranging from 0.15 to 0.25, to improve performance and thermal management. The heat sinks tested included straight, round, pin-finned heat sinks and copper foam heat sinks with a heat flux of 0.98 kW/m^2 , along with NePCM containing 0.25 wt% Al_2O_3 nanoparticles. The results showed that the base

temperature was reduced by 21.3 %, 25.03 %, and 36.2 %, respectively.

In cooling of batteries applications, Al-Rashed [272] developed a numerical study whereby hybrid nanofluid and NePCM were used to cool a lithium-ion battery that housed cylindrical cells. The finite element approach is used to conduct the simulations. According to the data, nanoparticle suspensions in PCM enhance thermal regulation. As a result, the PCM's melting and solidification times are decreased. A typical battery pack was used to examine the thermal protection performance of macro/nano-enhanced organic PCM (RT-44) for low to high discharge rates in the experimental work by Temel et al. [273]. Similarly, under the same conditions, NePCM increases the effective thermal protection time of the battery by 32 %–117 %, varying with the discharge rates. Consequently, reducing the cooling time for the next usage is the primary outcome of NePCM's contribution. NePCM into buildings is a new technology with great potential for minimizing thermal energy penetration into buildings, and green buildings improve indoor thermal comfort with minimal energy usage.

Ashok et al. [274] conducted an experimental work in which NePCM composite was directly poured into the matrix of a solid concrete hollow block of size $400 \text{ mm} \times 200 \text{ mm} \times 100 \text{ mm}$. These concrete blocks are used to build a one-meter cube experimental building to study the thermal implications of using this NePCM in buildings. Their findings proved that the NePCM-integrated building had a lower temperature with a delta of 5.3°C to 1.1°C than the building without NePCM. More recently, Li et al. [275] conducted an experimental analysis of the impact of nanoparticle migration on the thermal conductivity of NePCM in multilayer glazing, which is applied to windows and roofs. Thickness and tilt angle, which are the design parameters of NePCM, can be optimized using energy consumption and temperature differences between indoors and outdoors. The findings of their studies showed that NePCM's photoconversion efficiency and thermal conductivity are sensitive to the internal nanofluid movement and the cavity geometry. In addition, transverse NePCM has a relatively even temperature distribution across the width of the plate. According to the simulation results, NePCM with a 30° tilt angle is the most suitable for thermal performance improvement, and the thermal performance of a building can also be improved by using thicker windows and thinner roofs. There has been research conducted on NePCM in the context of renewable energy systems, such as PV.

According to Abdollahi and Rahimi [276], the photovoltaic panel's passive cooling mechanism makes use of natural water circulation. This is discussed in the article. As shown in Fig. 9, the PCM container, which had a zigzag shape that was specifically designed for it, was able to effectively reject undesired heat from the cooling water by increasing the area that was available for heat transfer.

The primary component of the PCM was a composite oil that was composed of 82 wt% coconut oil and 18 wt% sunflower oil throughout

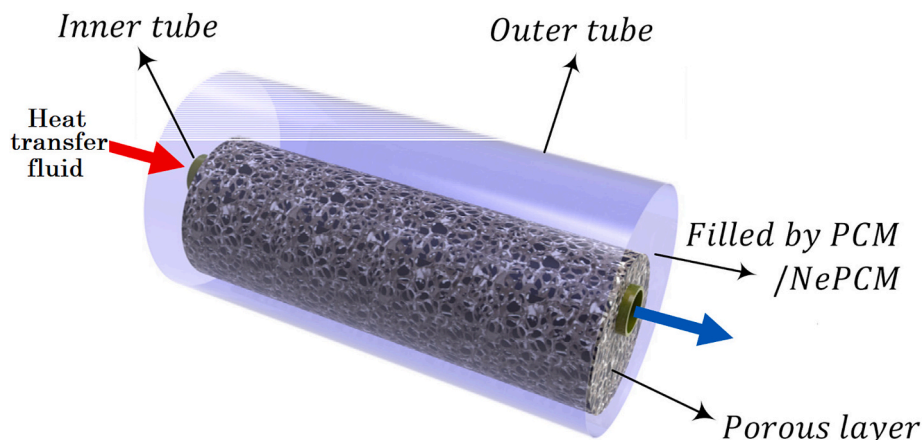


Fig. 8. Shell-and-tube reinforced with copper foam and PCM/NePCM [270].

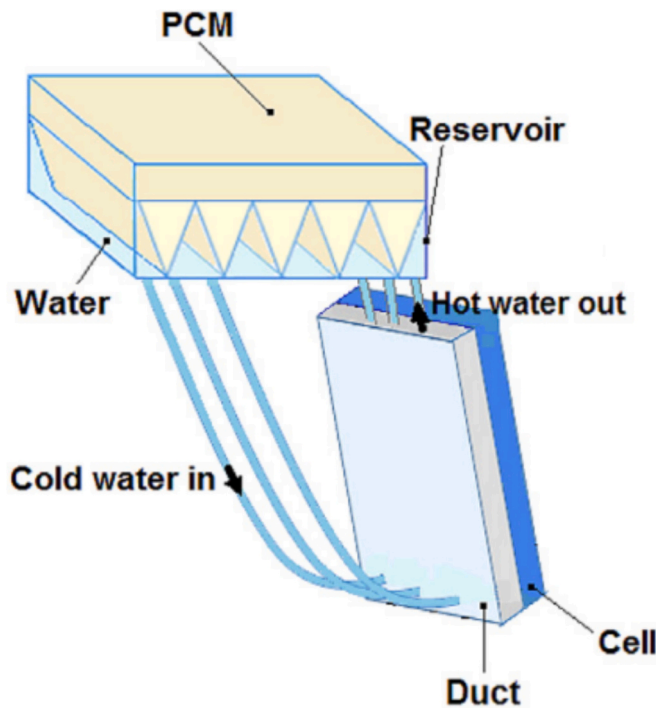


Fig. 9. Schematic diagram of the experimental rig [276].

the initial round of experiments. During the succeeding round of experiments, the composite oil was created by integrating Boehmite nanopowder as NePCM at a concentration of 0.9 wt% throughout the formulation process. In comparison to the PCM, the NePCM exhibited significantly better thermal performance on the test. The maximum power output showed the greatest increase for NePCM, which was reported at 44.74 %, 46.63 %, and 48.23 % after being exposed to insulation intensities of 410, 530, and 690 W/m², respectively. An experimental evaluation of a FPSC that utilised paraffin wax as a sustainable phase-change material was carried out by Bharathiraja and colleagues [277].

For the purpose of conducting studies that involve hybrid nanoparticles, namely MWCNTs and SiO₂ at different weight fractions, this study makes use of paraffin wax as the conducting matrix. The evaluation of water heating was carried out over eight hours, beginning at 8:00 AM and ending at 6:00 PM, using a tank that held 100 L of water. In comparison to the situation in which NePCM was not present, the implementation of NePCM in a f FPSC resulted in improved thermal performance as well as a marginal gain in thermal efficiency of roughly 7 %. With NePCM, the highest water temperature to be measured at the discharge was 70 °C. It is now clear that the technology of incorporating nanoparticles into PCMs enhances the charging and discharging performance of latent heat storage units, which enhances the thermal performance and thermal management of applications, with different enhancement rates depending on several factors, including the type of thermal application, configurations, type of PCM, nanoparticles, ambient conditions, heat transfer area, etc.

6.2. Energy: recent AI developments in energy optimization

AI can greatly enhance energy efficiency by optimizing operations, improving energy forecasting, and enhancing control strategies. AI can use advanced algorithms to analyse complex data patterns, optimize energy storage and release processes, and adapt thermal system parameters in real-time. It also accurately predicts energy demand and enables real-time monitoring and adaptive control, ultimately enhancing the efficiency and reliability of the power system. For the

purpose of predicting the performance parameters of desiccant-coated heat exchangers, Priyadarshi et al. [278] make use of ANFISs that are equipped with ANN. These heat exchangers have been constructed to make use of the fuzzy logic of ANN. Through the use of adaptive neuro-fuzzy and ANN, it was demonstrated that the proper design/inlet condition for the theoretical and design operating ranges was established.

When Abdul Hussein and Shbailat [279] conducted an experimental and numerical investigation, they used an ANN to estimate the thermal performance of a counter-flow double-tube heat exchanger that utilised bumpers to increase thermal performance. Also, they found that the thermal performance parameters of the building were precisely anticipated, which meant that they agreed with the results of the experimental tests. The bumpers are positioned between 130 and 190 cm away from the heat exchanger, which results in the best thermal performance possible. In their study [280], Askar and colleagues used ANNs to make predictions about the cooling and heating load of a building. Additionally, they presented the thermal performance of a four-layer wall in four distinct directions. As shown in Fig. 10 below, they decided to use MLP and radial basis networks in their neural network models. These networks were equipped with three different methods. Scaled conjugate gradient, Levenberg-Marquardt, and RBF are the three concepts. In addition, they created the data on 624 different configurations by utilizing six different building shapes, four different directions, five different glass areas, and five different glass distribution methods. According to their findings, the Levenberg-Marquardt algorithm provided the highest accuracy compared to experimental data. Also, it stated that for controlling the light and air, the window placement should be east and south with 20 % window area, and the position of the south-facing window will provide the nearest average heating and cooling loads.

For this study, Issa et al. [281] generated 432 models of three types of walls, light, medium, and heavy, which were made from different materials and thicknesses with the help of HAP software; the direction and colour of the wall from the outside were also taken into consideration. To predict the wall type that permits minimal external heat gains in hot climates, the data was used to train and test the ANN, whose architecture is illustrated in Fig. 11. A transient thermal analysis by Ansys was also applied to ensure the results' validity. For the optimal wall, the heat gain was 9.548, 9.598, and 9.62 W/m² for ANN, HAP, and Ansys, respectively, further validating the results.

Khan et al. [282] introduce a paper discussing the enhancement of the building heat load prediction through the methods considering the challenges of applying sophisticated control measures and AI integration to the air conditioning systems. The study presented an intelligent model of the predictive ANFIS, which gave a very high level of accurate prediction of HVAC performance outcomes and contributed to the generation of a holistic dataset. Ventilation depends on the weather parameters that affect convection, and they proposed the best set of weather conditions to be used: the wind speed of 5 km/h, relative humidity of 45 %, and the dry air temperature of 13 °C. The authors of a study by Rocha et al. [283] came up with a solution to the problem of energy planning in buildings by introducing the idea of structures that have a net energy cost that is zero. This is accomplished with the assistance of AI tactics, which include decision trees, binary gravity search algorithms, and bidirectional long short-term memories, among others.

Wind and photovoltaic (PV) renewable energies, an automatic capacitor bank to rectify the power factor, and a battery bank are the components that together make up the system, which is based on a genuine public building. It was determined that the distributed generating system had an output of 2.805 GWH per year, which was based on the fact that the building had an annual energy consumption of 1.748 GWH. The fact that the system is able to successfully cover about 160.5 % of the depot's electricity demands and 99.999 % of the energy costs that are incurred during the year is evidence that the idea of zero net energy is in existence. A payback period of 6.79 years was likewise proven by the system. Cox et al. [284] presented an integrated ANN and GA system to operate as a control system to lower operating costs in a

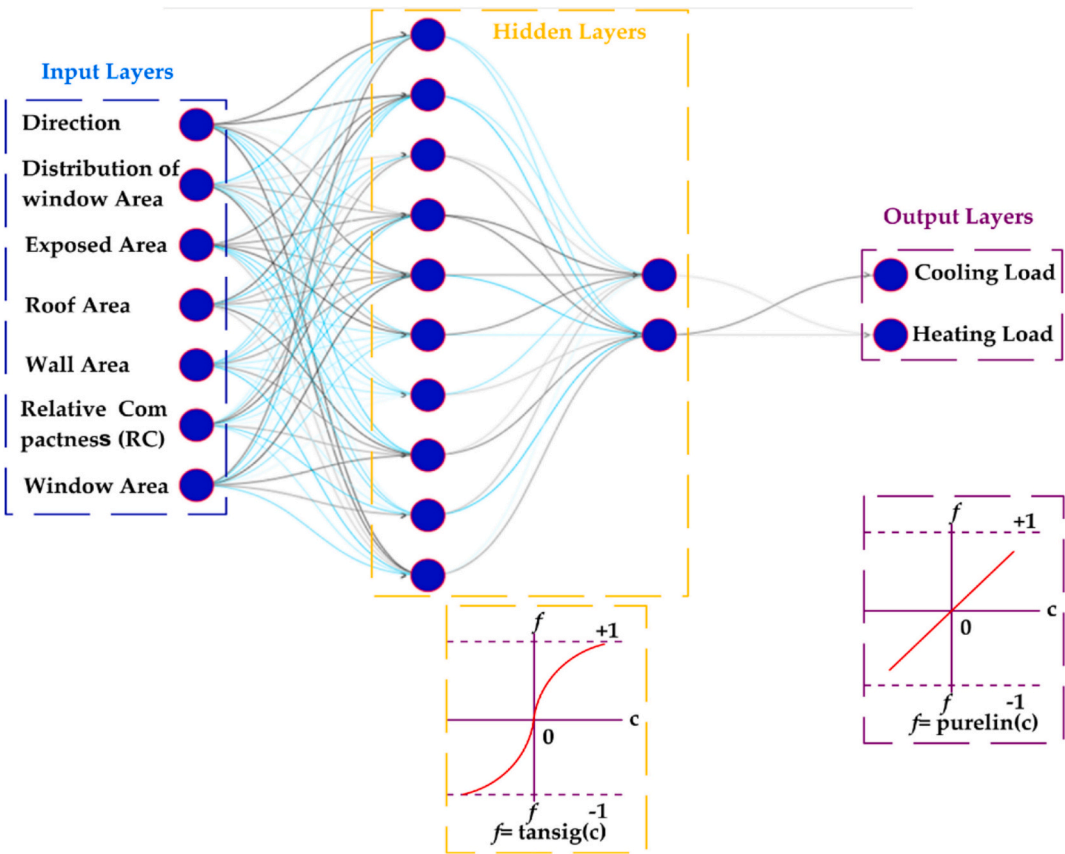


Fig. 10. Schematic diagram to predict cooling and heating loads by ANN [280].

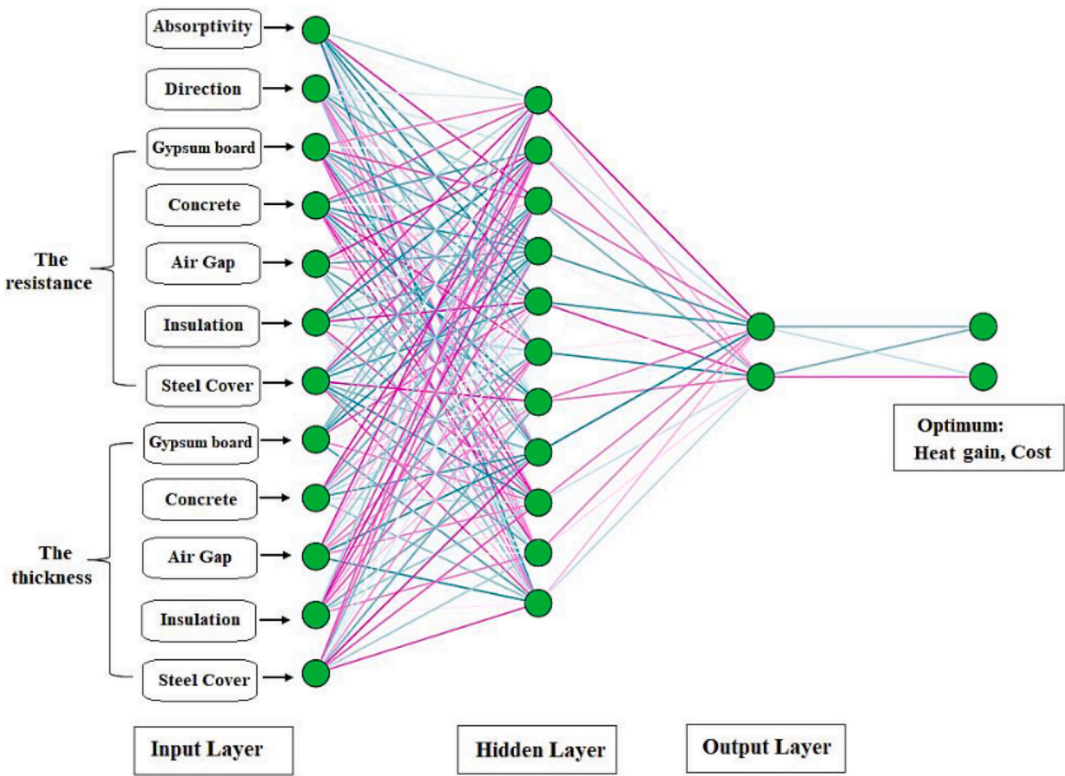


Fig. 11. Structure of ANN model used in [281].

district cooling system with ice storage. This system would take into consideration a variety of pricing situations. Other studies focused on other energetic and thermal applications with the same objective. Normalized mean bias error and RMSE were two of the metrics utilised in evaluating the model.

When attempting to forecast heat loads, various weather characteristics, such as the temperature of the dry bulb and the relative humidity, were gathered and compared with this information. The ANN-GA model attained a normalized mean bias error of 13.2 % and a cumulative variation of RMSE of 18.0 % on average. To forecast the amount of heat, load a vehicle will experience over a whole year, Askar et al. [285] created an ANNs model. Based on theoretical data, the ANN was trained to consider the heat generated in the cabin by passengers, sun radiation, the engine, ventilation, ambient, and exhaust environments. A maximum inaccuracy of 3 %, 5 %, and 11 % was found between the theoretically estimated heat load and the ANN for August, December, and September, respectively, according to the findings of their investigation. An ANN was used by Eldokaish et al. [286] in a PCM material to predict the phase transition length of encapsulated paraffin wax. The variables include volume percentages of manganese monoxide and silver nanoparticles, Hartmann numbers, and different strengths of radially dispersed magnetic fields. The training, testing, and validation sets' respective MSE values of 1.311, 1.628, and 2.329 demonstrate the noteworthy success that was attained.

Cui et al. [287] used an ANN to forecast and assess power output in a nanofluid/PCM photovoltaic TES. The solar panel temperature, global temperature, ambient temperature, and solar radiation were included in the input parameters, and the system produced the voltage, current, and power outputs. The ANN displayed excellent prediction performance, as shown in Fig. 12.

Leng et al. [288] used a genetic algorithm-based hybrid virus colony search on a water storage tank to evaluate best scheduling practices for a

hybrid power system incorporating a combined cooling, heating, and power. Their results indicated a summertime performance gain of 1.93 % overall and 1.92 % compared to other approaches. Lee et al. [289] utilised an ANN that was optimized by metaheuristics on a sensible chiller tank for latent heat thermal storage. In addition to lowering operational expenses by 9.1–14.6 %, they demonstrated an outstanding capacity to control load levels in a customisable manner. In order to forecast the phase transition process of PCM within a multi-channel thermal storage tank, Xiao et al. [290] developed a short-term memory (LSTM)-BP ANN. The temperature, the amount of time, and the velocity of the heat transfer fluid from the inlet are the most important characteristics that this ANNs takes into consideration. The findings indicate remarkable accuracy for the proposed model, achieving R-values of 0.99978 for the fluid ratio and 0.99921 for temperature. In order to detect the presence of a leak and the location of any leak in a water testing platform, Mahdi et al. [291] constructed an ANN model.

The parameters, neurons, and hidden layers of this model were optimized during the process. In this particular instance, the data from the accelerometers and dynamic pressure sensors that were collected in the time domain were processed into higher levels of statistics, such as the autocorrelation coefficient and signal energy. They were able to achieve maximum accuracies and F1 scores of 86.5 % and 86.2 %, respectively, with their method, which enables reasonably reliable leak identification and localisation across a variety of situations using their method. The results of this study highlight the potential for water management systems to incorporate such advanced features and combinations of AI. Predictive control models, especially those based on ANN, show effective control capabilities and provide promising and concrete solutions that are difficult to implement by conventional experimental and numerical methods.

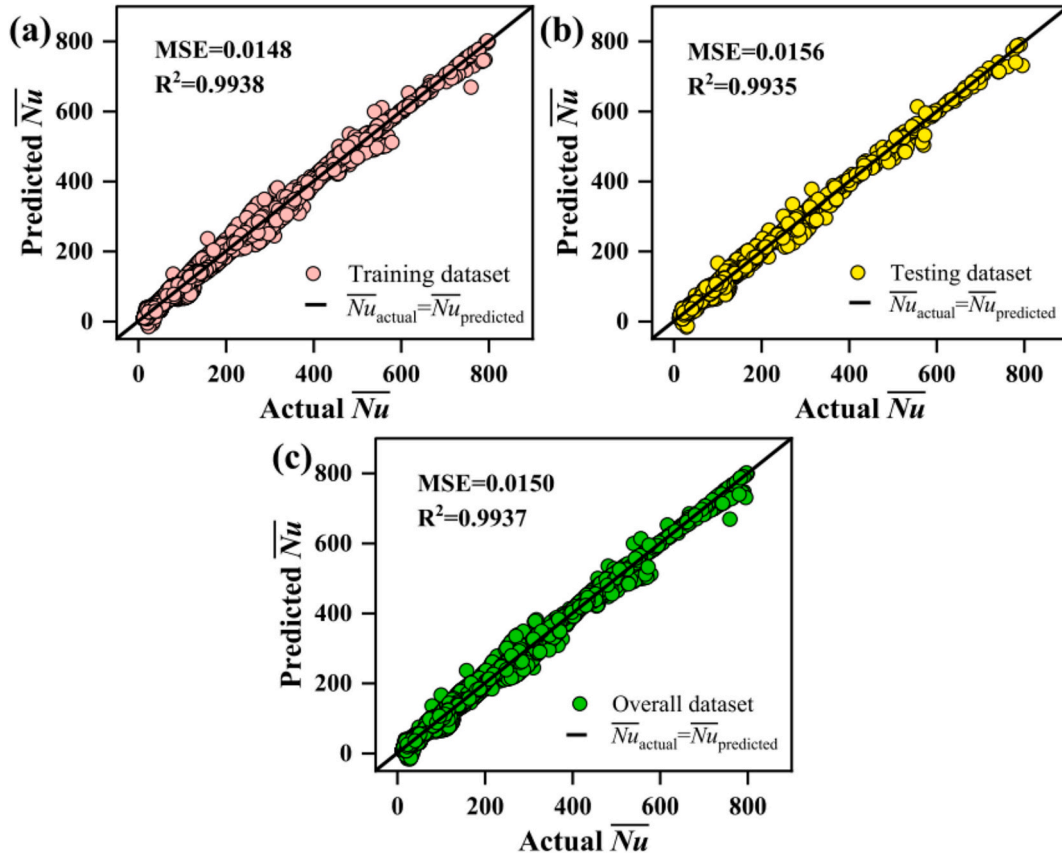


Fig. 12. Verifying the average predictive Nusselt number in case (a) training, (b) testing, and (c) overall datasets [287].

6.3. Case studies: successful applications of AI and nanotechnology

This section explores some case studies of AI and nanotechnology in improving application performance, highlighting successful implementations, lessons learned, best practices, and measurable benefits. Siemens has created an AI-driven energy management system designed for buildings that enhances the efficiency of HVAC systems by considering occupancy trends and weather conditions. This system has been implemented in various buildings, leading to notable energy savings. In its data centres, Google utilised AI to improve cooling systems. By evaluating sensor information and weather predictions, the AI cut cooling energy use by as much as 40 %. Similarly, Microsoft has adopted AI technology in its data centres to optimize energy consumption. By accessing data from sensors and weather forecasts, AI algorithms modify cooling and power usage, reducing energy consumption while ensuring performance remains intact. The quality of data sourced from sensors and other inputs is vital for optimizing performance through AI. Integrating data from diverse sources and verifying its accuracy and dependability [292]. Moreover, energy performance driven by AI can positively influence the environment by decreasing greenhouse gas emissions and other pollutants linked to energy production [293].

Nanotechnology has achieved remarkable success in various thermal applications; for example, in a double-pipe heat exchanger, Kadhim et al. [294] improved its thermal performance by using Fe_3O_4 -water nanofluid as a heat transfer fluid instead of water, where the Nusselt number increased by 13 %, which means that the size of the heat exchanger can be reduced to achieve the same required performance. The hybrid MgO - CuO -water nanofluid was able to increase the average thermal efficiency of a heat pipe evacuated solar water collector by about 20 % compared to using pure water, as reported by Aytaç et al. [295], which led to economic improvements and reduced the recovery period by about 27 %. Razzaq et al. [296] incorporated TiO_2 nanoparticles with mineral oil lubricant at different ratios to improve the performance of an air conditioner compressor. They found that lubricants containing 0.4 % TiO_2 nanoparticles had lower energy destruction and power consumption, which improved the system economically as well as provided environmental benefits. Abdullah et al. [297] coated a solar still with black-enhanced CuO nanoparticles to enhance the heat transfer properties between the basin surfaces and the water. Their results indicated a 14 % improvement in freshwater productivity with the coating. In the same study, they also mixed these nanoparticles with PCM to further enhance the performance of the solar still, and indeed, the productivity increased by 108 %. The various studies chosen in distinct thermal applications offer a significant understanding of the extensive advantages provided by nanotechnology and AI. These technologies can contribute to cost savings for organizations by minimizing energy usage. Among the financial advantages of AI-driven applications are decreased energy expenses and lower maintenance costs.

Beyond theoretical advances, commercial viability is a decisive factor in the successful adoption of AI- and nanotechnology-based TESs. Several industrial-scale implementations provide encouraging evidence. For instance, a pilot project by the Fraunhofer Institute in Germany combined nano-enhanced PCMs with predictive AI control to manage building HVAC systems in a smart campus setting. Over a 12-month deployment, the system demonstrated a 28 % reduction in energy consumption and a 21 % improvement in occupant comfort, with a return on investment (ROI) of under 3 years. Similarly, Tesla's Gigafactory in Nevada integrates AI-optimized thermal management for its battery production line using AI algorithms trained on both ambient and operational data. The system enables dynamic control over heat exchangers and coolant flows, contributing to energy savings of up to 15 % annually.

The design and implementation of a multi-physics simulation environment to explore the synergistic design of AI and nanofluid-based thermal systems was done in a recent theoretical study by Mohib Husain et al. (2025) [298]. The researchers used a computational fluid

dynamics (CFD) model with ML-based surrogate modelling to forecast and optimize the thermal performance of a nanofluid-cooled FPSC at different operating conditions. The nanofluid was a hybrid Al_2O_3 - Cu /water mixture with different volume fractions of 0 % to 2 %. Its authors have created a set of CFD simulations, which span a variety of Reynolds numbers, solar intensities, and concentrations of nanoparticles. The CFD results were then used to train an ANN to match the CFD results closely. ANN gave a MAPE of less than 2.5, which shows very good correlation with the numerical model. As it was discovered in the study, the thermal efficiency increased by 17.6 % against conventional water, with the optimum concentration of nanoparticle at 1.2 %. In addition, the computational cost was reduced considerably as the optimization process took place more than 85 % faster via the trained AI model. These findings demonstrate the potential of AI to supersede theoretical modelling to facilitate faster parametric studies and improve the dynamic process of designing high-performance thermal systems. It also demonstrates a convenient approach to hybrid AI-CFD modelling, which may also be used in future studies in which nanotechnology is applied in the field of heat transfer.

However, these deployments also revealed practical constraints that must be considered in future commercial analyses. Challenges include the high upfront costs of nanomaterial integration, the need for specialized hardware for AI computation, and regulatory barriers in energy infrastructure upgrades. Additionally, while AI models offer superior performance in data-rich environments, their efficacy may degrade in data-scarce or unstable deployment contexts. Thus, future commercial strategies should not only rely on performance metrics but also consider lifecycle cost analysis, scalability under variable climate zones, and long-term reliability. Including such real-world feedback loops into system design and investment planning is essential for transitioning from experimental validation to widespread industrial adoption.

6.4. Future trends in research and development

Edge AI is an emerging trend along with IoT, highlighted as AIoT, proposing an improved energy efficiency performance in various sectors like buildings, industrial applications, and operational performance in transportation networks. The AI algorithms are being used to analyse real-time data about energy consumption to minimize wastage and bring down operational costs. However, as AI and nanotechnology merge into thermodynamics, sustainability and environmental impacts must be taken up comprehensively. Future work should prioritize the subsequent components:

- **Life Cycle Assessment:** Life cycle assessment of nanomaterials deployed within TESs is important for assessing their environmental impact from cradle-to-grave scenarios. LCA can provide insight into resource consumption, energy use, and emissions related to the use of nanomaterials, elucidating against possible environmental trade-offs and benefits.
- **Sustainable Material Development:** Research should also include the development of sustainable nanomaterials that reduces adverse environmental impacts. This includes efforts to explore biodegradable or less toxic alternatives to conventional nanoparticles, along with the search for greener synthesis processes that lower energy use and waste production.
- **Integration of Sustainability Metrics:** Henceforth, AI models should integrate sustainability metrics in their optimization algorithms. While AI works toward the design of more sustainable TESs, parallel to the global sustainability goals, it would do so by assessing environmental impacts along with performance metrics.
- **Regulatory Frameworks:** Increased use of nanotechnology in energy systems would call for the establishment of a clear regulatory framework. These studies should assist in framing environment

health-related policy measures for safe and sustainable utilization of nanomaterials.

- **Standardized Testing Protocols:** Future research ought to be centered on the development of standardized means of testing and criteria of performance threshold of the nanomaterial-based enhanced thermal systems. This would make it reproducible, cross-comparable and would allow the adoption of nanotechnology in energy systems at industrial scales.
- **AI Training Datasets:** Datasets that can be used to train ML models in thermal energy better are severely lacking, particularly publicly accessible, topic-specific datasets. Open data would foster benchmarking ability, accountability, and a fair performance assessment of different competing AI algorithms.

In addressing these issues through a critical stance, future research can underpin claims of sustainable advancement in the converged domains of nanotechnology and AI in thermal applications. This will make a case for technologies whose energy efficiency is counterpoised with environmental sustainability and public health.

7. Challenges and future directions

There are some **challenges** and issues regarding the use of nanotechnology that must be addressed, which can be summarized as follows:

- Additional research is required to investigate the environmental effects of environmental particles, with a particular emphasis on the disposal of discarded nanomaterials and any potential long-term repercussions.
- Serious consideration must be given to the cost of manufacturing nanoparticles and the application of these particles on an industrial scale.
- When it comes to nanofluids, the high-pressure drop that is linked with nanofluid flow is a significant cause for concern. However, while utilizing nanoparticles, it is necessary to take into consideration their viscosity. There is also a significant worry regarding the stability of nanofluids over lengthy periods of operation. Nanoparticles can aggregate, which can result in contamination of equipment and a decrease in the efficiency of heat transfer.
- Using nanoparticles to improve thermal conductivity in latent thermal storage materials may cause issues such as inhomogeneous temperature distribution during charging and discharging. Further research is needed to address these issues.

When it comes to AI, it is confronted with a variety of significant issues, some of which are technological in character, such as those in the energy industry; others are industrial and regulatory; and still others are related to user awareness [299]. The following is a list of the numerous difficulties that still need to be addressed by the application of AI:

- **Absence of adequate finances:** For AI technology to assimilate more widely and deeply into the energy system, we require a bundle of AI workers who wield appropriate technology to instigate this change, but this is wanting the change in the AI-led energy market. Furthermore, this technological deployment in the energy market includes the creation, optimization, and operation of heavily resource- and financially intensive programs. A consistent shortage of developed financial expertise is one of the most pressing challenges for integrating intelligent energy system development.
- **Smart integration:** One of the key challenges in deploying smart production technology involves integrating old and new equipment. Also, some consider old equipment unwieldy, while others see it as more of a challenge than an opportunity.
- **Smart manufacturing security issues:** The smart manufacturing process uses a network system that interlinks production units and a

manufacturing unit to various end users in an industrial context. So that networking systems such as smart production processes can be developed securely.

- **Implementation of AI:** Like with all technologies, the biggest challenge that stakeholders and energy companies face is adapting to AI, which will take time, dollars, and effort.
- **Computing strength:** The continuous development of algorithms with AI requires advanced computers in terms of cores and graphics processors to work with the required performance. Therefore, you need computing strength that meets the requirements, and these super-computers are not cheap.
- **Data Protection:** Data protection has become one of the major challenges of the century, and with the continuous development of AI, data protection has become a natural challenge for AI, especially in the field of energy to all cyber-attacks.

Future research in AI for thermal application performance optimization should involve interdisciplinary collaboration between AI researchers, thermal energy experts, policymakers, and industry stakeholders. This collaboration can lead to innovative solutions and best practices. Data sharing and standardization are crucial for the success of AI in process optimization. Also, the potential of integrating nanotechnology and AI into thermal applications is yielding amazing results that open up new societal horizons.

Future research should also address the practical integration challenges between AI models and physical thermal systems, such as sensor drift, latency, and hardware compatibility, which can impair real-time control and system stability. Developing standardized validation protocols (spanning both simulation and experimental data) is essential to ensure reproducibility and cross-study comparability. Moreover, the use of AI-driven material discovery frameworks, including RL and inverse design, holds strong potential for optimizing nanoparticle properties and accelerating the development of high-performance, sustainable nanomaterials.

8. Conclusion

This review illustrates that integrating nanotechnology and AI may significantly enhance TESs' performance, efficiency, and sustainability. Nanotechnology, with advancements in nano-enhanced materials such as PCMs and nanofluids, tackles significant issues in thermal conductivity and energy storage, improving system efficiency in applications including solar collectors and energy storage systems. These materials enhance thermal management by expediting heat transfer rates and augmenting energy storage capacity, demonstrating particular significance in renewable energy systems where efficiency and dependability are paramount. Conversely, AI enhances these systems via predictive maintenance, real-time performance monitoring, and energy forecasting. Algorithms like ANNs and optimization methods provide intelligent control, reducing energy losses and enhancing operational stability.

Notwithstanding these achievements, concerns remain regarding economic viability, material expenses, and the environmental sustainability of nanomaterials. Future research should emphasize the creation of economical, eco-friendly materials and the enhancement of sophisticated AI algorithms capable of adapting to the intricacies of TESs.

Furthermore, enhancing AI skills to augment the scalability and integration of thermal systems inside extensive energy networks is crucial for wider deployment. Integrating nanotechnology with AI is a viable avenue for developing efficient, robust, and sustainable thermal energy solutions. This convergence facilitates the shift to renewable energy sources and corresponds with global energy sustainability objectives, representing a substantial advancement in the evolution of optimized intelligent energy systems.

CRediT authorship contribution statement

Hayder I. Mohammed: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Farhan Lafta Rashid:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Hussein Togun:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Ephraim Bonah Agyekum:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Arman Ameen:** Writing – review & editing, Visualization, Project administration, Methodology, Formal analysis. **Karrar A. Hammoodi:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Rujda Parveen:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Saif Ali Kadhim:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Walaa N. Abbas:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization.

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Data availability

No data was used for the research described in the article.

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